

The Trial

Methodology & Operations Lead · CLARION-3

15 DAYS · 61 STOPS · GRADE 12 · THE_TRIAL

THE OPENING CARD

CLARION-3 has been running for four years: 2,400 people across thirty-one hospitals, half of them getting a new drug and half the usual treatment. Nobody in this building knows which patient got which. You are the methodology and operations lead at the coordinating centre, so the pack the independent monitoring board reads is the pack you assemble. That board sits in three weeks, and it can stop the trial. Stop it early on a result that turns out to be noise, and a drug that works is buried for a decade. Carry on past the point the evidence was in, and every patient who joined after today was in a trial that already had its answer.

What the trial already decided

BEFORE THE DAY — GREET

One floor, four functions, and none of them may tell the others everything

The coordinating centre is one long floor: statistics at one end, the monitors in the middle, safety and the sites at the other. Feldman, the trial statistician, wants you known to as many of them as you can before the first interim, because the firewall means the person you have to ask is often the one person who cannot tell you why they are asking back.

PLAN CARD 110W

Monday, and the second interim analysis is 21 days away. CLARION-3 has randomised 2,400 people across 31 hospitals, and in 3 weeks Professor Vogt's board will read the trial by arm and say whether it carries on. Yemi Balogun wants that meeting to end the trial; Miriam Feldman wants it to end nothing that was not written down in advance. Today you go back through what the trial committed to before the first participant — the assignment, the outcome it counts, the size it was built to detect — because everything the board is told in three weeks is measured against those promises rather than against what anybody now hopes.

BRIEFING 12W

Three weeks to the second interim, and the board pack starts today.

WHAT YOU DECIDE 9W

Establish what CLARION-3 promised before it had any data.

1.1 Concealment is not the same as blinding

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A ROOM · CHOICE

SCENE 37W

Farida Diouf has a note from site 7. Their pharmacy diary lists kit numbers beside the arm each one contains, and the research nurse can see the next entry before she decides whether to enter a patient.

GUIDE 65W

All four options are real costs of a readable diary. Ask of each whether it acts at the moment of entry, or afterwards. A nurse who can see the next slot can wait a day for a patient she thinks will do well. Nobody has to be dishonest for that to wreck the comparison. Concealment protects entry; blinding protects everything after, and they fail separately.

ASK 8W

What has site 7's readable diary cost CLARION-3?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

participants are assigned to arms by chance rather than by choice

VERDICT 87W

Randomisation only buys balanced groups if nobody can see what is coming. A nurse who knows the next slot is the drug can wait a day for a patient she thinks will do well, or enter a frailer one when the next slot is standard care. Neither is dishonest and both destroy the comparison, because the arms now differ in who was entered as well as in what they were given. Concealment protects the moment of entry. Blinding protects everything after it, and the two fail separately.

TAKEAWAY 17W

If the next assignment can be seen in advance, chance is no longer the only thing choosing.

1.2 One outcome, named in advance

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A PERSON · STRESS

SCENE 33W

Kwame Ansah has four measurements from the same participant on his screen and a definition agreed four years ago. A journalist has asked the sponsor which of the 4 CLARION-3 is actually about.

GUIDE 63W

Four candidate primary endpoints, scored on two things the sponsor cares about. The slider is the event rate the trial actually gets, which nobody knows in advance — slide it to the pessimistic end and see which endpoint still answers the question there. The one that looks best at the expected rate is not necessarily the one that survives being wrong about it.

ASK 17W

Which primary endpoint still answers the question if the event rate comes in lower than anyone expected?

BEHIND THE BUTTON 170W

What a primary endpoint is. The one question the trial was built to answer, named before the first participant was randomised and used to set the sample size. It also fixes how the result will be reported: an absolute risk reduction in percentage points, with the relative figure beside it. Everything else on Ansah's screen is collected — worth having, worth reporting, and not what the trial promised.

Why naming it in advance matters. With four measurements per participant, somebody choosing afterwards has four chances to find a difference, and the reported result would be the best of four rather than the answer to one question. That is the whole of why the definition is four years old.

Why the event rate is the assumption that bites. Every endpoint's power depends on how often its event happens. A rare, hard endpoint is unimpeachable and may never accumulate enough events to say anything; a composite happens more often and answers a blurrier question. That trade is what the slider makes visible.

TAKES AS READ

a trial collects more than one measurement from each participant

VERDICT 164W

Naming the endpoint in advance fixes how the answer gets reported. It comes out as an absolute risk reduction in percentage points, with the relative figure beside it. Every endpoint here is defensible. They fail differently when the assumption behind them turns out to be wrong. Cardiovascular death alone asks the crispest question and needs the most events. At 3.5 per cent a year it is decisive. At 1.6 per cent the trial never accumulates enough, and reports an interval that includes everything. The composite of death, stroke or hospitalisation happens often enough to answer even at the low end. The cost is that it blurs what "the treatment worked" means. The walk distance would give a number at any event rate. It answers a different question

— how people felt, not what happened to them. So choosing well costs something. The composite is not the sharpest question on the list. It is the one still being asked when the trial's central assumption misses.

TAKEAWAY 19W

The primary endpoint is the one question the trial was built to answer; the rest are collected, not promised.

1.3 A trial ends at a number of events

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · BALLPARK

SCENE 32W

Balogun wants to know when CLARION-3 will be finished and has been told, twice, that nobody can say. Feldman writes three quantities on the whiteboard and asks him to estimate it himself.

GUIDE 61W

Five numbers, and three belong to other questions: the events adjudicated so far, the hospitals recruiting, and one of the two rates. Ask of each whether this schedule depends on it. A trial learns from events, not from people. So the duration is not a decision anybody makes. It is what the rate produces, and the rate belongs to the participants.

ASK 15W

Estimate how long CLARION-3 has to follow its participants to reach the events it needs.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the trial compares how often an event happens in two arms · events accumulate over time, at some rate per participant per year

VERDICT 118W

A time-to-event trial learns from events, not from people. 2000 participants who stay well carry almost no information about which arm is better; the comparison rests on the ones something happened to. So the design fixes a number of events, and recruitment and follow-up run until that number arrives. How long that takes is not a decision anybody makes — it is the required events divided by the rate they arrive at, and that rate belongs to the participants rather than to the trial. A healthier-than-expected group is bad news for the schedule for exactly this reason, and it is why the interim looks are placed at fractions of the events rather than at fractions of the years.

TAKEAWAY 16W

A trial that waits for events finishes when the events arrive, which is not a date.

END OF THE DAY 12W

A trial's claims are the ones it made before it had data.

Who measures, and how

BEFORE THE DAY — TRIAL

Walk the floor, statistics to sites

The distance down this floor is distance from the patient, and it is the whole geography of the trial. Walk it once. A query that starts at the site desk and ends in statistics passes four rooms, and knowing which room is which is how you avoid sending a question to the only people who must not see the answer.

PLAN CARD 111W

Tuesday. 20 days to the interim, and the board pack has 2 outcomes in it rather than 1. CLARION-3 counts a clinical event, which a committee rules on from the notes, and it also carries a symptom score the treating doctor writes down at each visit. Kwame Ansah trusts the first and says so at every meeting; Nuno Iriarte at site 12 thinks the second is the 1 that matters to patients, and Maggie Doyle agrees with him. Today you work out where each number is actually made, and what each is worth when the two arms are compared — because in three weeks they will be on the same page.

BRIEFING 14W

Two outcomes, two ways of measuring, and only one of them has a committee.

WHAT YOU DECIDE 13W

Find out where a number is made before deciding what it is worth.

2.1 A number can be precise and still be the wrong outcome

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · PROTOCOL

SCENE 40W

Kwame Ansah has 4 CLARION-3 measurements on his table. One comes from a national register, one from blinded readers, one from a treating doctor, and one from a central laboratory. Doyle wants to know what can go wrong with each.

GUIDE 61W

Four measurements and four weaknesses. Pair them by asking how much trial judgement each measurement passes through. A death in a national register goes through almost none. A borderline admission goes through a definition. A symptom score goes through a person who may believe something. And one of them can be measured perfectly and still stand in for the wrong thing.

ASK 12W

Match each measurement to the weakness that can bend what it means.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a measurement can be precise without being clinically important · a surrogate outcome is a measure used to stand in for a clinical outcome

VERDICT 79W

Measurements can fail in different ways. A death in a national register needs little trial judgement. A borderline admission can move if readers apply the definition differently, so blinded adjudication protects it. A symptom score can move with the scorer's belief about treatment. A laboratory marker can be measured perfectly and still mislead. If it stands in for a clinical outcome, it is a surrogate. Changing the marker does not prove that admissions, symptoms or survival changed with it.

TAKEAWAY 20W

Before trusting a number, ask both how it was made and whether it measures the outcome people actually care about.

2.2 What counts as a deviation

SITE OPERATIONS · MONITORS' ROOM · A PERSON · CHOICE

SCENE 32W

Lucia Marchetti has four things sites did this month that were not what the protocol says. None of them harmed anybody. Site 12 has asked which of them are worth the paperwork.

GUIDE 62W

All four are real deviations and none harmed anybody. Ask of each whether it lands on both arms equally. A late blood test and a late form are untidy in both arms. A different pump delivers the same drug. Only one of these changes who is in the trial. It matters because those entries need not be spread evenly across the arms.

ASK 10W

Which of these most threatens the comparison CLARION-3 is making?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the protocol states what should happen at each visit

VERDICT 86W

Deviations matter in proportion to what they do to the two arms. A late blood test and a late form are untidy and affect both arms equally, so the comparison survives them. A different pump delivers the same drug. Entering people the trial was not designed for changes who is in it — and if that site's entries are not spread evenly across the arms, it changes what is being compared. The test is always whether the two groups are still alike apart from the treatment.

TAKEAWAY 17W

A deviation is recorded for what it does to the comparison, not for how bad it looks.

2.3 The plan that was in force

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · SEQUENCE

SCENE 35W

Nkechi Umeh has four versions of the analysis plan, each dated. Feldman wants to know which 1 governs the interim in 20 days, and the sponsor's lawyer wants to know how anybody could prove it.

GUIDE 65W

All four steps will happen, so ask what order makes the claim provable.

Prespecification is a claim about time: this analysis was chosen before anybody could see what it would give. The only support for that afterwards is a dated public record made before the data existed. Reverse any two of these and the arithmetic is still right about a question chosen after the fact.

ASK 14W

Order what has to be true before an interim analysis can be called prespecified.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a trial's plan can be amended while it is running

VERDICT 96W

Prespecification is a claim about time: this analysis was chosen before anybody could see what it would give. The only way to support that claim afterwards is a public record with dates on it, made before the data existed. That is why the plan is registered rather than filed, why the cut comes after the registration rather than before, and why the person who runs it is the one person already permitted to see the arms. Reverse any two of those steps and the analysis is still correct arithmetic about a question chosen after the fact.

TAKEAWAY 14W

A result is worth more when the plan that produced it can be dated.

A measurement is worth what the procedure that made it is worth.

The signal at two sites

BEFORE THE DAY — FOLLOW

A monitoring visit, walked once at the pace it happens

Source data verification is not reading a file — it is standing beside somebody while they find the original, and the visit has a rhythm. Stay with the monitor. The pauses are where the discrepancies get noticed, and somebody hurrying the visit finds nothing.

PLAN CARD 94W

Wednesday morning, 19 days out. Sites 4 and 19 have each reported a participant admitted with the same unusual liver picture, 11 days apart, and neither report says which arm the participant was on because neither site knows. Feldman's instinct is that 2 is a number you see in a trial this size and that the board meets in 19 days anyway. Balogun wants it in front of Vogt this week. Today you decide what is reported, to whom, and how fast — and what the trial is entitled to do while it waits.

BRIEFING 13W

Two sites have sent the same kind of event in the same fortnight.

WHAT YOU DECIDE 13W

Decide what has to be reported today rather than at the next meeting.

3.1 The clock that is not the analysis clock

SAFETY MONITORING · MONITORING BOARD ROOM · A ROOM · TRIAGE

SCENE 35W

The second admission arrived 18 minutes ago. 1 patient is still on the ward, and the 2 sites are 400 kilometres apart. Vogt is reachable this afternoon. The next board meeting is 19 days away.

GUIDE 57W

Four options, and two clocks are running. One patient is still on the ward and the board meets in nineteen days. Ask of each option which clock it obeys. A serious unexpected event may change treatment tomorrow. Whether it also counts as an endpoint is a separate path, and the same event can travel both at once.

ASK 8W

What should happen to these two reports today?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

serious harm during a trial is reported whether or not the drug is thought to have caused it · hard endpoints against judged ones, and blinded adjudication — taken as read

VERDICT 90W

Safety reporting and endpoint analysis run on different clocks. A serious unexpected event may matter to treatment tomorrow, so it cannot wait 19 days for a board meeting. The site reports what happened now, with enough detail to judge urgency. Whether the event later counts toward a trial endpoint is a separate process. The same event can therefore travel down two paths at once: immediate safety reporting and later blinded adjudication. Waiting for one path to finish before starting the other loses time the safety system was built to protect.

TAKEAWAY 13W

A safety report does not wait for the analysis it might belong to.

3.2 Ten minutes on each case

SITE OPERATIONS • MONITORS' ROOM • A PERSON • VALUE

SCENE 38W

Lucia Marchetti has ten minutes with each site before the reports go to Vogt. One coordinator is at a bedside. Five useful questions are on her screen, but asking all of them would take longer than either call.

GUIDE 59W

Ten minutes with each site, five useful questions, and no time for all of them. Open each question and ask what its answer would change before the reports go to Vogt. One coordinator is at a bedside, so the call has to be short and worth having. Spend the ten minutes on what the reports cannot already tell you.

ASK 13W

What should Marchetti spend the ten minutes finding out before the reports leave?

BEHIND THE BUTTON 132W

Why some questions are wasted on a call. Anything the site has already entered in the system is in the reports. A call is the only way to get at what has not been written down: why a visit slipped, what the coordinator is worried about, whether a protocol step is being interpreted differently.

Why 'before the reports leave' is the constraint. Once the report is out, a correction is a deviation and a conversation. Something learned now changes what the report says; the same thing learned tomorrow changes what has to be explained.

What a coordinator at a bedside can actually do. Answer one or two things from memory. A call that asks five questions gets three rushed answers, and the rushed ones are the least reliable data in the trial.

TAKES AS READ

a safety pattern depends on whether cases share timing and clinical features • a VALUE board charges time for each piece of evidence • eligibility criteria, and what widening them changes — taken as read

VERDICT 79W

Two reports become a signal only when their details line up in a way that could matter. Timing is the fastest discriminator here. A case three days after first exposure tells a different story from one eight months later. Matching intervals in two distant sites would strengthen the case for a shared cause. Different intervals would weaken it. Investigator opinions, recruitment totals and withdrawal status may still matter, but none can establish that temporal pattern before the reports leave.

TAKEAWAY 19W

When time is scarce, collect the evidence that can change the decision before collecting detail that only decorates it.

3.3 What has to be told, and how fast

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · PROTOCOL

SCENE 29W

Umeh has a wall calendar with two windows marked on it in red and a stack of four events from this month. She has to sort them before Friday.

GUIDE 61W

Four events and four obligations. Pair them by asking three things of each: is it serious, was it unexpected, and could it be related? The combination sets the clock, not how dramatic it sounds. A death nobody thinks is related is still serious. And an admission for the trial's own condition is the thing being counted rather than a safety report.

ASK 10W

Match each event to what the trial owes and when.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

regulators set deadlines for reporting harm during a trial · hard endpoints against judged ones, and blinded adjudication — taken as read

VERDICT 86W

The reporting window is set by the category, not by the drama. Serious, unexpected and possibly related is the combination that runs on a clock of days, because it is the combination that might change how the drug is used tomorrow. A death nobody thinks is related is still serious and still reported, on the ordinary schedule. An admission for the trial's own condition is the thing being counted, so it goes to adjudication rather than to the safety desk. Everything else lives in the file.

TAKEAWAY 16W

Expedited reporting is decided by what the event is, not by what anybody thinks caused it.

3.4 Concealment is not the same as blinding

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A ROOM ·
CALLBACK · CHOICE

SCENE 37W

Farida Diouf has a note from site 7. Their pharmacy diary lists kit numbers beside the arm each one contains, and the research nurse can see the next entry before she decides whether to enter a patient.

GUIDE 65W

All four options are real costs of a readable diary. Ask of each whether it acts at the moment of entry, or afterwards. A nurse who can see the next slot can wait a day for a patient she thinks will do well. Nobody has to be dishonest for that to wreck the comparison. Concealment protects entry; blinding protects everything after, and they fail separately.

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Randomisation only buys balanced groups if nobody can see what is coming. A nurse who knows the next slot is the drug can wait a day for a patient she thinks will do well, or enter a frailer one when the next slot is standard care. Neither is dishonest and both destroy the comparison, because the arms now differ in who was entered as well as in what they were given. Concealment protects the moment of entry. Blinding protects everything after it, and the two fail separately.

TAKEAWAY 17W

If the next assignment can be seen in advance, chance is no longer the only thing choosing.

END OF THE DAY 10W

Safety reporting runs on a different clock from the analysis.

The unplanned look

PLAN CARD 99W

Thursday, 18 days out. The sponsor's medical director has asked, in writing, for a pooled summary of the primary endpoint by arm before the board meets, on the grounds that the liver reports have made everybody nervous. Feldman has said no twice and been asked to explain why in a way that does not sound like obstruction. Alina Petrescu, who is the only person who could produce it, has said nothing at all. Today you work out what an extra look actually costs the trial, who may take one, and what the trial can offer instead of the answer.

BRIEFING 14W

Somebody senior would like a peek at the numbers before the board sees them.

WHAT YOU DECIDE 12W

Establish what looking early costs and who is allowed to do it.

4.1 What an extra look spends

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · BALLPARK

SCENE 38W

The medical director wants one unplanned endpoint summary before noon and has already drafted the email asking for it. Feldman has the two planned looks on the whiteboard and an empty box where this new request would sit.

GUIDE 61W

Add up what the planned looks already spend, and compare it with what the trial has to spend in total. Each look has a boundary, and the error rate it uses is the price of that look. Pick the three planned amounts. The unplanned look's price is on the board too, and it is what the sum has no room for.

ASK 12W

What do the planned looks already spend of the trial's error budget?

BEHIND THE BUTTON 167W

What is being spent. A five per cent error rate is a promise about the whole trial, not about any one test. Every look at the accumulated data is another chance to catch a wandering difference at its most flattering moment, so each look consumes part of that budget and the budget does not refill.

How planned looks are paid for. The boundaries at the planned looks are made harder — an interim result has to be more extreme than the final one to stop the trial — and the amounts add up to no more than the total. That is the whole of the arithmetic: the spending plan is written before any data arrive.

Why an unplanned look is different in kind. It has no boundary and no allowance, so whatever it spends comes out of the final analysis. A trial that has spent more than its budget can no longer say what its error rate is, which is a worse position than not having looked.

TAKES AS READ

a trial accepts a small chance of calling a difference real when there is none · type I and type II error, and what α and β are for — taken as read

VERDICT 156W

The rule is $\Sigma \alpha_{\text{spent}} \leq \alpha$. This trial is run one-sided at 0.025, and the spending plan puts 0.001 at the first interim, 0.004 at the second and 0.020 at the final analysis: $0.001 + 0.004 + 0.020 = 0.025$, which is the whole budget and exactly what the plan is allowed to be. That is why the interim boundaries are so hard to cross — an early look is cheap precisely because it demands an extreme result. An unplanned look priced at 0.015 takes the total to 0.040, and the trial can no longer state its own error rate: 4 per cent of trials with no real effect would report one. Nothing about the data changes when somebody looks, which is what makes this counter-intuitive. What changes is the number of chances the trial has taken to be fooled, and that number is in the denominator of every claim it goes on to make.

TAKEAWAY 14W

A trial's tolerance for error is spent by testing, and it does not refill.

4.2 Why the board looks and the team does not

SAFETY MONITORING · MONITORING BOARD ROOM · A PERSON · CHOICE

SCENE 34W

Vogt is in the building for an hour. Petrescu's closed report is on the table, face down. The medical director is copied on everything the trial team writes and on nothing the board writes.

GUIDE 54W

Four options, and somebody has to see the arms or a harmful treatment continues unnoticed. Ask of each whether it names a route from knowledge into the trial's own conduct. The board does not recruit, treat or score. The team does all three. Confidentiality cannot close those routes, and being advisory does not either.

ASK 14W

Why is the monitoring board shown the arms when the trial team is not?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the trial team recruits, treats and measures while the trial runs

VERDICT 75W

Somebody must see unblinded data or a harmful treatment could continue unnoticed. The safeguard is separation of roles. The monitoring board does not recruit participants, deliver treatment or score outcomes. What it learns cannot change those processes directly. The trial team does all three. If that team knows which arm is winning, ordinary judgement can shift enrolment or follow-up. It can also shift how an ambiguous score is written. Confidentiality alone cannot remove those pathways.

TAKEAWAY 16W

The people who could act on an early result are the people kept from seeing it.

4.3 When the blind is opened for one person

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A ROOM · SEQUENCE

SCENE 37W

Diouf has a call from an intensive care registrar at three in the morning, on speaker. A CLARION-3 participant is in front of him and the treatment he is considering depends on what she has been receiving.

GUIDE 55W

Four steps, and the rail is not a clock — every one of them could be done in the next ten minutes. Order them by what each one protects: the thing that cannot be recovered if it is skipped goes first, and the thing that can be done afterwards without costing anybody anything goes last.

ASK 13W

Order the four steps by what is lost if the step is skipped.

BEHIND THE BUTTON 170W

Why the blind exists. It protects the comparison, not the participant. Every person who knows an assignment is a person whose judgement can drift with it — the treating team, the assessor, the participant. The trial's answer depends on that ignorance being intact across four hundred people, and one code being opened costs one of them.

Why the clinical decision comes first anyway. The blind is a research instrument and this is a patient in front of a registrar at three in the morning. If knowing the assignment changes what is safe to give her, the code is opened; the trial can lose a participant from the blinded analysis and cannot lose her.

What has to be written down and why the order matters. The reason, the time and who authorised it, recorded before the envelope is opened rather than reconstructed afterwards — because once the assignment is known, nobody can prove the request was made on clinical grounds. Telling the monitoring committee can wait; telling the site file cannot.

TAKES AS READ

each participant's assignment is held in a sealed record that can be opened · blinding, and the several people it can fail for separately — taken as read

VERDICT 139W

The first step cannot be recovered, because it cannot be established after the fact. Once the registrar knows the assignment, nobody can show the request rested on clinical need rather than curiosity. The written reason is the same problem in another form. A note made before the envelope is opened is evidence. One made afterwards is an account. Only then is the code opened, and the participant treated on what is known. Notifying the committee and flagging her in the analysis plan matters, and costs nothing by waiting until morning. That is why it is last rather than unimportant. The blind protects the comparison across four hundred people. It

is opened for one of them when a decision for that person turns on it. The price is paid in the analysis, not argued about at three in the morning.

TAKEAWAY 17W

The blind exists for the trial; it is opened when a decision for one person needs it.

4.4 One outcome, named in advance

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · CALLBACK · STRESS

SCENE 33W

Kwame Ansah has four measurements from the same participant on his screen and a definition agreed four years ago. A journalist has asked the sponsor which of the 4 CLARION-3 is actually about.

GUIDE 63W

Four candidate primary endpoints, scored on two things the sponsor cares about. The slider is the event rate the trial actually gets, which nobody knows in advance — slide it to the pessimistic end and see which endpoint still answers the question there. The one that looks best at the expected rate is not necessarily the one that survives being wrong about it.

ASK 17W

Which primary endpoint still answers the question if the event rate comes in lower than anyone expected?

BEHIND THE BUTTON 170W

What a primary endpoint is. The one question the trial was built to answer, named before the first participant was randomised and used to set the sample size. It also fixes how the result will be reported: an absolute risk reduction in percentage points, with the relative figure beside it. Everything else on Ansah's screen is collected — worth having, worth reporting, and not what the trial promised.

Why naming it in advance matters. With four measurements per participant, somebody choosing afterwards has four chances to find a difference, and the reported result would be the best of four rather than the answer to one question. That is the whole of why the definition is four years old.

Why the event rate is the assumption that bites. Every endpoint's power depends on how often its event happens. A rare, hard endpoint is unimpeachable and may never accumulate enough events to say anything; a composite happens more often and answers a blurrier question. That trade is what the slider makes visible.

TAKES AS READ

a trial collects more than one measurement from each participant

VERDICT 164W

Naming the endpoint in advance fixes how the answer gets reported. It comes out as an absolute risk reduction in percentage points, with the relative figure beside it. Every endpoint here is defensible. They fail differently when the assumption behind them turns out to be wrong. Cardiovascular death alone asks the crispest question and needs the most events. At 3.5 per cent a year it is decisive. At 1.6 per cent the trial never accumulates enough, and reports an interval that includes everything. The composite of death, stroke or hospitalisation happens often enough to answer even at the low end. The cost is that it blurs what "the treatment worked" means. The

walk distance would give a number at any event rate. It answers a different question — how people felt, not what happened to them. So choosing well costs something. The composite is not the sharpest question on the list. It is the one still being asked when the trial's central assumption misses.

TAKEAWAY 19W

The primary endpoint is the one question the trial was built to answer; the rest are collected, not promised.

END OF THE DAY 14W

A trial's small budget for being wrong is spent by looking, not by finding.

The site that is faster than everybody

PLAN CARD 100W

Friday, 17 days out. Site 12 has randomised more than twice its expected share and Nuno Iriarte has been thanked publicly for it twice. Marchetti has now put his screening log beside the other 30 and found that where most sites screen 4 people to enter 1, site 12 enters almost everyone it sees. Balogun reads that as a site that has organised itself well. Today you work out which of those readings the log actually supports, and what it would mean for the trial if the fast site is also entering a different kind of participant from everybody else.

BRIEFING 10W

One site's screening log is shorter than every other site's.

WHAT YOU DECIDE 12W

Work out why site 12 recruits twice as fast as anyone else.

5.1 What a short screening log says

SITE OPERATIONS • MONITORS' ROOM • A ROOM • CHOICE

SCENE 30W

31 screening logs on 1 page. Most sites screen four and enter one. Site 12 screens 11 and enters 9, and its consent rate is the highest in the trial.

GUIDE 57W

Four explanations for a short screening log. Ask of each what the log would look like if it were true. The high entry rate is not the strongest clue. The nearly empty reasons column is. Eleven screened and nine entered should still record why two were not, and a year should show many who failed before consent.

ASK 8W

Which explanation best fits site 12's screening log?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

sites record everybody they assess as well as everybody they enter • eligibility criteria, and what widening them changes — taken as read

VERDICT 77W

The high entry rate is not the strongest clue. The nearly empty reasons column is. A site that truly screened 11 people and entered 9 should still record why the other 2 were not entered. Across a year, it should also show many people who failed before consent. If those people never reach the log, the trial cannot describe who was considered and passed over. The result is a selected group whose recruitment funnel cannot be reconstructed.

TAKEAWAY 14W

Recruitment figures describe a funnel, and an empty log hides the top of it.

5.2 Balance is bought in blocks

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A ROOM · CHOICE

SCENE 33W

Diouf pulls site 12's kit records. The site has entered 94 participants, 49 to 1 arm and 45 to the other, in a sequence that never runs more than 4 the same way.

GUIDE 60W

Four purposes for a short block. Ask of each whether it is bought or paid for. Chance is fine over two thousand people and unreliable over ninety. Blocks force the counts to catch up every few entries. Predictability is the price rather than the point, which is why the block length varies and nobody at the site is told it.

ASK 16W

Why is the sequence built so that no run of the same arm goes past four?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

assignment is by chance, in a sequence prepared before the trial opened

VERDICT 88W

Pure chance is fine over 2000 people and unreliable over 90. A site that closes at 19 participants could easily have 13 on 1 arm and 6 on the other, and any difference between sites then partly becomes a difference between arms. Working in small blocks forces the counts to catch up every few entries, so the arms are balanced within each site at every point. The cost is predictability, which is why the block length varies and why nobody at the site is told what it is.

TAKEAWAY 14W

Chance alone is unreliable in small numbers, which is why the sequence is constrained.

5.3 Make the three-point offset move

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · CONTROL

SCENE 38W

Kwame Ansah has matched visit records from site 12 and 3 comparison sites. The symptom score sits about 3 points higher at site 12. A re-scoring console can change one feature of the setup for the same records.

GUIDE 66W

The number you are watching is the offset in symptom score between site 12 and the comparison sites, currently about three points. The console can change one feature of how the same records were produced, and the run-to-run noise is about half a point, so anything smaller than that is not a response. Change one, run it, restore it, and name the feature the offset follows.

ASK 15W

Which single change makes the offset disappear, and does putting it back restore the offset?

BEHIND THE BUTTON 166W

Why the records are held fixed. Nothing about the patients changes in this console: the same visits, the same people, the same dates. Only a feature of how the score was produced changes. That is what makes it a controlled comparison rather than a new study, and it is why a response points at the procedure rather than at the disease.

What the three candidates are for. Who does the scoring, how visit days were matched, and what mix of cases each site saw are three ordinary ways a site can look different without being different. Each is a real explanation somebody at the meeting will offer, and only one of them survives being switched.

Why half a point is the floor. Rescoring the same records twice does not give the same number twice — scales have judgement in them. That scatter is the yardstick every response has to beat, and a difference under it is a measurement of the yardstick rather than of the site.

TAKES AS READ

a systematic offset is a repeated shift rather than random scatter · a control test changes one factor while holding the rest fixed · hard endpoints against judged ones, and blinded adjudication — taken as read

VERDICT 83W

A site-wide offset can come from the patients, the timing of visits or the person applying the scale. The hard endpoint already argues against a healthier site 12 population. The control test goes further. It keeps the same records and changes one candidate cause at a time. Only the reader change moves the score by far more than the background noise. When the original reader is restored, the offset returns. That reversal is much stronger evidence than a correlation between site and score.

TAKEAWAY 20W

A suspected cause becomes persuasive when changing only that factor moves the reading and reversing it brings the effect back.

5.4 Eleven a week, through the summer

SITE OPERATIONS • MONITORS' ROOM • A PERSON • HOLD

SCENE 39W

The trial needs eleven entries a week to finish on time. Sites go on holiday, a competing trial opens at the largest one, and the only lever the office has is where it puts its monitors and its money.

GUIDE 53W

Hold the entry rate inside the band on the tracker. The band narrows as the summer goes on, because the shortfall that could be made up in September cannot be made up in November. Site support is your control, and every change at a site keeps costing entries until the support answers it.

ASK 11W

Hold the entry rate at eleven a week through the summer.

BEHIND THE BUTTON 121W

Why a rate rather than a total. Recruitment is a funnel — screened, eligible, consented, entered — and each stage has its own losses. A site that stops screening does not lose one patient; it loses its share of every week that follows.

Why a shortfall compounds. The trial ends when its events accumulate, and events accumulate from patients already in it. A patient not entered in July is not contributing follow-up in August either, so a week's shortfall costs more than a week.

Why the band tightens. Early in the summer there is time to recover; by the end there is not. The same rate below target is a larger fraction of what remains recoverable with every week that passes.

TAKES AS READ

a trial enters patients at some rate over time • eligibility criteria, and what widening them changes — taken as read

VERDICT 155W

Recruitment is a rate and every disturbance to it is a rate too. A site's principal investigator going on leave does not cost one patient; it costs that site's weekly share for as long as the leave lasts. A competing trial at the largest site takes a fraction of every week's screening from then on. A new site opening adds a fraction the same way. None of them is a step, which is why moving a monitor for one week and moving them back changes nothing that lasts. And the shortfall compounds in a way a simple total would not show: an event-driven trial finishes when its events accumulate, and events come from patients already enrolled, so a patient not entered in July is also not contributing follow-up in August. That is why the band narrows through the summer. The same rate below target is recoverable in June and is the trial's timeline in September.

TAKEAWAY 14W

A shortfall in a rate is not made up by a good week later.

5.5 A trial ends at a number of events

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · CALLBACK · BALLPARK

SCENE 32W

Balogun wants to know when CLARION-3 will be finished and has been told, twice, that nobody can say. Feldman writes three quantities on the whiteboard and asks him to estimate it himself.

GUIDE 61W

Five numbers, and three belong to other questions: the events adjudicated so far, the hospitals recruiting, and one of the two rates. Ask of each whether this schedule depends on it. A trial learns from events, not from people. So the duration is not a decision anybody makes. It is what the rate produces, and the rate belongs to the participants.

ASK 15W

Estimate how long CLARION-3 has to follow its participants to reach the events it needs.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the trial compares how often an event happens in two arms · events accumulate over time, at some rate per participant per year

VERDICT 118W

A time-to-event trial learns from events, not from people. 2000 participants who stay well carry almost no information about which arm is better; the comparison rests on the ones something happened to. So the design fixes a number of events, and recruitment and follow-up run until that number arrives. How long that takes is not a decision anybody makes — it is the required events divided by the rate they arrive at, and that rate belongs to the participants rather than to the trial. A healthier-than-expected group is bad news for the schedule for exactly this reason, and it is why the interim looks are placed at fractions of the events rather than at fractions of the years.

TAKEAWAY 16W

A trial that waits for events finishes when the events arrive, which is not a date.

END OF THE DAY 9W

A site cannot see what is unusual about itself.

The amendment comes back

BEFORE THE DAY — HUNT

Six unblinding envelopes, and the safe holds four

Every unblinding envelope has to be accounted for before the committee meets, and two are not in the safe. An envelope somebody could have opened is a broken blind whether or not anybody opened it, and the trial is defended on the record rather than on goodwill.

PLAN CARD 102W

Monday, 14 days out. The amendment signed eight months ago to keep recruitment on schedule widened two entry criteria, and it worked: the trial filled. The data manager, Tobias Renner, has now plotted events against months and the curve has flattened since. Feldman thinks the trial is further from its answer than the calendar says; Balogun signed the amendment and says the alternative was a trial that never finished at all. Today you work out how much information CLARION-3 actually holds, what the flattening does to the interim in a fortnight, and whether the 2 of them are disagreeing about anything real.

BRIEFING 7W

Recruitment recovered. The event rate did not.

WHAT YOU DECIDE 12W

Find out what widening the entry criteria has done to the trial.

6.1 How much of the trial has happened

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · BALLPARK

SCENE 34W

Feldman has two numbers on the board: the events CLARION-3 was designed to need, and the events adjudicated so far. Balogun has a third, the months elapsed, and thinks it tells the same story.

GUIDE 59W

Five numbers, and three of them measure calendar time or people: the months elapsed, the months planned, and the number randomised. Ask of each whether an interim boundary depends on it. A trial three quarters through its years can hold half its events. That is why the meeting moves when the event curve flattens, and the date does not.

ASK 11W

Estimate the fraction of CLARION-3's information that is in hand today.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the trial is powered on a number of events rather than a number of people

VERDICT 93W

Calendar time and information are different quantities, and only one of them decides what an interim can say. A trial three quarters of the way through its planned years may hold half its events, because the people recruited late have been followed for months rather than years and because a healthier group produces events more slowly. The boundary at each interim is set by the information fraction, not by the date, so a flattening event curve moves the meeting without moving the calendar. It is also why finishing takes longer than anyone forecasts.

TAKEAWAY 17W

The share of the planned events already collected is what says how far through a trial is.

6.2 Who a wider door lets in

SITE OPERATIONS · MONITORS' ROOM · A ROOM · BALLPARK

SCENE 32W

Marchetti has the entry criteria as they were and as they are, with the participants recruited under each. Balogun wants to know whether the amendment cost anything he should have seen coming.

GUIDE 69W

The amendment brought in people at lower risk, so the hazard ratio the trial can expect moved from 0.75 toward 0.85. Work out how many events the trial now needs to answer its question at the same power. The numerator is the same for both; what changes is the squared logarithm underneath. One tile is that term for the original criteria, which is the comparison rather than the answer.

ASK 16W

How many events does the trial need now the wider criteria have diluted the expected effect?

BEHIND THE BUTTON 154W

Why events and not people. An event-driven trial is powered by how many events it observes, not by how many participants it enrolls. Recruiting lower-risk people adds participants who contribute little to that count, which is why a wider door can grow the trial and slow it down at the same time.

Why the effect size is squared underneath. The number of events needed goes as one over the square of the log hazard ratio, so a treatment effect that dilutes a little costs a lot. Moving from 0.75 to 0.85 roughly triples the events required, which is the amendment's real price.

What the amendment did not do. It did not make the treatment work less well. It changed who is in the trial, and therefore the average effect the trial is measuring and how often its event happens. Both belong in the sentence, and only one of them is usually said out loud.

TAKES AS READ

entry criteria decide who may be randomised

VERDICT 160W

The event requirement is $\text{events} \approx 4 (z_{\alpha/2} + z_{\beta})^2 / (\ln \text{HR})^2$. At 5 per cent two-sided and 80 per cent power the z terms are 1.96 and 0.84, so the numerator is $4 \times 2.80^2 \approx 31.4$ and it does not change. Under the original criteria the expected hazard ratio was 0.75, so $(\ln 0.75)^2 = 0.0828$ and the trial needed $31.4 \div 0.0828 \approx 380$ events. The amendment admitted lower-risk participants whose expected benefit is smaller, moving the trial-wide hazard ratio to about 0.85: $(\ln 0.85)^2 = 0.0264$, and $31.4 \div 0.0264 \approx 1,190$ events. Three times the information for the same question, because the effect size sits squared in the denominator. And the second cost compounds it: lower-risk participants have events less often, so each new enrolment

contributes less to the count the trial is waiting for. The amendment grew the trial and slowed it down, and neither effect was on the form that authorised it.

TAKEAWAY 16W

Widening entry changes who is in the trial, and therefore how often the trial's event happens.

6.3 Changing the plan, on the record

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · SEQUENCE

SCENE 30W

Umeh has the amendment file open: the change, the approvals, the registry update and the date each happened. One of the four is a fortnight later than it should be.

GUIDE 60W

All four steps will happen, and one of them is a fortnight late. Ask of each what order makes the change defensible. Approved by outsiders, published before use, then handed to sites. That sequence shows the change was not chosen after seeing what it would do. Enrol first and register after, and nothing can ever show the two were independent.

ASK 12W

Order what has to happen before a widened criterion may enrol anybody.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

the entry criteria are part of the protocol and can be amended

VERDICT 82W

The order is the protection. A change made for a stated reason, approved by people outside the trial, published before it is used and only then handed to sites, is a change nobody can have chosen after seeing what it would do to the results. Enrol first and register afterwards and there is no way, ever, to show the two were independent. The audit does not ask whether widening was wise — it asks whether the record was made before it mattered.

TAKEAWAY 19W

An amendment is judged by whether it was recorded before it was used, not by whether it was sensible.

6.4 Eligible, or nearly

SITE OPERATIONS · MONITORS' ROOM · A PERSON · BELT

SCENE 34W

The screening log has more entries than the meeting has time for. Some of these people meet the protocol as written. The rest are close, and close is not a category the protocol has.

GUIDE 40W

Two bins. Somebody is eligible if every criterion in the protocol is satisfied as it is written. Anything that requires reading a criterion generously is not eligible, however sensible the generosity looks. Sort on the protocol, not on the patient.

ASK 14W

Send each screening entry to the bin that says whether the protocol admits them.

BEHIND THE BUTTON 80W

Why the line is where it is. Every criterion was written to make the trial's population the one the question is about. A patient admitted by a generous reading is in the analysis but not in the population, and nothing downstream can separate them again.

Why "nearly" is the dangerous pile. Nobody enrolls somebody wildly ineligible. Trials are damaged by a hundred small generousities, each defensible on its own, which together make a population nobody planned and no protocol describes.

TAKES AS READ

a protocol states who may be entered into a trial

VERDICT 152W

Eligibility criteria are what make a trial's population the population its question is about, and they only work if they are applied as written. The dangerous entries are never the obviously ineligible ones — nobody enrolls a patient twenty years outside the age range. They are the ones a sensible person would let through: an eGFR of 29 against a limit of 30, an ejection fraction measured six weeks ago instead of four, a diagnosis made clinically rather than by the stated test. Each is defensible alone. A hundred of them together produce a trial population nobody planned, whose result applies to nobody in particular, and the analysis cannot separate them out afterwards because the log records that they were eligible. That is why the honest handling of a near-miss is a protocol amendment, prospectively, with the reason written down — and not a decision at the screening desk on a Thursday.

TAKEAWAY 12W

A criterion read generously is a criterion the trial no longer has.

6.5 A number can be precise and still be the wrong outcome

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · CALLBACK · PROTOCOL

SCENE 40W

Kwame Ansah has 4 CLARION-3 measurements on his table. One comes from a national register, one from blinded readers, one from a treating doctor, and one from a central laboratory. Doyle wants to know what can go wrong with each.

GUIDE 61W

Four measurements and four weaknesses. Pair them by asking how much trial judgement each measurement passes through. A death in a national register goes through almost none. A borderline admission goes through a definition. A symptom score goes through a person who may believe something. And one of them can be measured perfectly and still stand in for the wrong thing.

ASK 12W

Match each measurement to the weakness that can bend what it means.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a measurement can be precise without being clinically important · a surrogate outcome is a measure used to stand in for a clinical outcome

VERDICT 79W

Measurements can fail in different ways. A death in a national register needs little trial judgement. A borderline admission can move if readers apply the definition differently, so blinded adjudication protects it. A symptom score can move with the scorer's belief about treatment. A laboratory marker can be measured perfectly and still mislead. If it stands in for a clinical outcome, it is a surrogate. Changing the marker does not prove that admissions, symptoms or survival changed with it.

TAKEAWAY 20W

Before trusting a number, ask both how it was made and whether it measures the outcome people actually care about.

END OF THE DAY 17W

A trial's progress is measured in events, and a decision made a year ago can slow them.

Who is missing

PLAN CARD 114W

Tuesday, 13 days out. Renner's reconciliation says 91 participants have no twelve-month outcome: some moved, some withdrew, some stopped answering the phone. Ansah cannot adjudicate what nobody wrote down, and 48 of the 91 stopped attending in the weeks after a worsening episode was recorded in their notes. Doyle has asked, at the steering committee, what the trial intends to say about people it lost, and whether the answer changes depending on which arm they were on. Today you decide how the missing are handled in the analysis the board reads in 13 days, what the sites should spend a fortnight chasing, and what CLARION-3 can honestly claim about the people it cannot find.

BRIEFING 13W

91 people have no outcome recorded, and they are not a random 91.

WHAT YOU DECIDE 13W

Decide what to do about the participants the trial has lost track of.

7.1 Missing is not one thing

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · PROTOCOL

SCENE 39W

Friday's database lock is now on the wall clock. Four kinds of missing record are still in the queue, each with a count beside it. Renner wants one rule for all of them before the next site call starts.

GUIDE 54W

Four gaps and four consequences. Pair them by asking whether the reason for the gap is connected to how the person was doing. A move abroad is not. Stopping attendance after worsening is. A lost fax batch is recoverable. And one of them closes the door permanently, which is a different problem from either.

ASK 10W

Match each gap to what it does to the comparison.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an outcome can be absent for reasons connected to the outcome itself

VERDICT 86W

Missing data are dangerous for different reasons. A missed visit because somebody moved is mostly a loss of precision. A missing outcome after a participant worsened is different, because the reason for the gap is tied to the thing being measured. That can bend the comparison. A form still sitting at a site is recoverable and should be chased before lock. Withdrawal of consent changes what the trial is allowed to pursue. One rule for every blank cell would hide these differences instead of managing them.

TAKEAWAY 17W

A gap that depends on what happened is not the same as a gap that does not.

7.2 Everyone as assigned

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A PERSON · BALLPARK

SCENE 29W

Two analyses are on Feldman's desk. One counts everybody in the arm they were assigned. The other counts only those who completed treatment, and it shows a bigger effect.

GUIDE 62W

Work the relative risk out for the analysis the board is shown — everybody in the arm they were assigned to, whatever they went on to do. Pick the event rate in the treatment arm and the event rate in the control arm and divide. Two of the tiles are the per-protocol rates, which is the other analysis and the other answer.

ASK 11W

What is the relative risk in the analysis by assigned arm?

BEHIND THE BUTTON 168W

What relative risk is. The event rate in the treatment arm divided by the event rate in the control arm. Below one means fewer events on treatment. It says nothing about how many people that is — a relative risk of 0.7 can be six fewer events per hundred or six per hundred thousand, which is why it needs the absolute difference beside it.

Why as-assigned is the analysis of record. Randomisation made the arms alike in everything, including what nobody measured, and that balance belongs to the groups as assigned. Remove people for something that happened afterwards — stopping the drug, missing visits — and they are removed for a reason connected to how they were doing.

Why the smaller effect is the honest one. As-assigned usually shows less benefit, and that is not a defect of the method. It is the effect of being offered the treatment, which is the decision a doctor actually faces in a clinic where some patients will not finish the course.

TAKES AS READ

some participants do not take the treatment they were assigned

VERDICT 160W

Relative risk is $RR = p_{\text{treatment}} / p_{\text{control}}$. As assigned, 84 of 600 on treatment had the event and 120 of 600 on control did, so the rates are 0.140 and 0.200 and $RR = 0.70$ — a 30 per cent relative reduction. Counting only those who completed gives 0.121 against 0.201 and $RR = 0.60$, which looks better and is not the same question. The people dropped from that second analysis stopped treatment, and stopping is connected to how they were doing, so the two arms are no longer alike in the way randomisation made them. The as-assigned figure is the effect of being offered the treatment, which is what a clinic faces. And note what neither number says: a relative risk hides how big the effect is. The absolute difference here is six

events per hundred, so about 17 people have to be treated to prevent one — the honest headline, and it is not in the ratio.

TAKEAWAY 10W

Analysing people as assigned keeps the balance that randomisation bought.

7.3 40 hours before the lock

SITE OPERATIONS • MONITORS' ROOM • A ROOM • ALLOCATE

SCENE 38W

Lucia Marchetti has a fortnight, 31 sites and 40 monitoring hours she can still deploy. Four recoverable gaps compete for that time. 14 people who withdrew consent are locked out of the plan and cannot be contacted again.

GUIDE 59W

Forty monitoring hours, four recoverable gaps, and a fortnight before the lock.

Watch the questions list rather than the site map: it shows what the interim analysis can still answer under your plan. The fourteen who withdrew consent are locked out and cannot be contacted, so no plan recovers them — spend the hours on what is still recoverable.

ASK 11W

Which monitoring jobs buy the answers the interim analysis actually needs?

BEHIND THE BUTTON 122W

What makes a gap recoverable. A missing form that exists on paper, a visit whose data was never entered, a query nobody answered. Those can be closed by somebody going to look. A withdrawal cannot, because the data no longer belongs to the trial.

Why the interim analysis defines value. Not every gap affects it. A missing secondary endpoint at a small site may not move anything, while a handful of missing primary events at a large one can decide whether the trial continues.

Why hours rather than sites. Monitoring time is the resource, and sites differ enormously in how long a gap takes to close. Four hours at a well-run site can recover more than a day at one in difficulty.

TAKES AS READ

missing outcomes are most dangerous when the reason for missingness is tied to the outcome • withdrawal of consent prevents further contact

VERDICT 79W

The dangerous gaps are not simply the largest ones. 48 participants disappeared after worsening, so their missingness is tied to the outcome itself. That can bend the comparison if the losses differ by arm. Seven outcomes already exist on paper and are cheap to restore. People who moved are missing for a more neutral reason, so tracing them mainly improves precision. Rechecking complete records buys less. The allocation should protect validity first, then spend any remaining effort on precision.

TAKEAWAY 17W

A monitoring plan is a set of questions it can answer and questions it deliberately leaves unanswered.

7.4 The plan that was in force

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · CALLBACK · SEQUENCE

SCENE 35W

Nkechi Umeh has four versions of the analysis plan, each dated. Feldman wants to know which 1 governs the interim in 20 days, and the sponsor's lawyer wants to know how anybody could prove it.

GUIDE 65W

All four steps will happen, so ask what order makes the claim provable.

Prespecification is a claim about time: this analysis was chosen before anybody could see what it would give. The only support for that afterwards is a dated public record made before the data existed. Reverse any two of these and the arithmetic is still right about a question chosen after the fact.

ASK 14W

Order what has to be true before an interim analysis can be called prespecified.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a trial's plan can be amended while it is running

VERDICT 96W

Prespecification is a claim about time: this analysis was chosen before anybody could see what it would give. The only way to support that claim afterwards is a public record with dates on it, made before the data existed. That is why the plan is registered rather than filed, why the cut comes after the registration rather than before, and why the person who runs it is the one person already permitted to see the arms. Reverse any two of those steps and the analysis is still correct arithmetic about a question chosen after the fact.

TAKEAWAY 14W

A result is worth more when the plan that produced it can be dated.

What you do about missing outcomes is a decision, and it has to be made before you see them.

The cold room

PLAN CARD 99W

Wednesday, 12 days out. Site 19's fridge logger shows 41 hours above the storage range in July, and the alarm was acknowledged twice and never escalated. Farida Diouf has the kit numbers that were in that fridge and can say exactly who received them. What she cannot say, and will not find out, is which of those boxes held the drug. Today you decide what the excursion means for the participants who received that kit, what has to be reported and to whom, and how the trial keeps a warm fridge from becoming a reason to open the blind.

BRIEFING 13W

A freezer alarm at one site was silenced for two days in July.

WHAT YOU DECIDE 14W

Work out which kit is compromised and what that obliges the trial to do.

8.1 Kit that cannot be vouched for

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A ROOM · CHOICE

SCENE 36W

Site 19 has already dispensed 11 boxes from the warm fridge to 9 participants. Farida Diouf has the logger trace, dispensing list and stability sheets open. A replacement shipment will not reach the site until morning.

GUIDE 57W

Four readings of a warm fridge. Ask of each whether the fault can act on both arms equally. Identical boxes can hold different contents, and heat has nothing to damage in a comparator. So the direction of the error is not symmetric. Eleven boxes is a question about size, which is separate from the question about direction.

ASK 15W

What is the most important effect of the site 19 excursion on the trial comparison?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

both arms receive kit that looks identical

VERDICT 79W

Identical boxes can hide different contents. Heat can damage active drug, while the comparator has no active material to lose. The excursion can therefore pull treated participants toward the control result. That is different from an error that affects both arms equally. 11 boxes are unlikely to change the whole trial, but the scale of the problem is a separate question from its direction. Recording the event lets later reviewers distinguish a small one-sided fault from a repeated one.

TAKEAWAY 12W

A storage failure damages one arm's material and leaves the other untouched.

8.2 What the nine are owed

SITE OPERATIONS · MONITORS' ROOM · A PERSON · TRIAGE

SCENE 29W

Nine people received kit from that fridge. Marchetti has to tell site 19 what to say to them, and Doyle has asked to see the wording before it goes.

GUIDE 65W

Four things to tell nine people, and two duties pulling opposite ways. Ask of each whether it can be said to all nine equally without saying what is in the box. Saying they may have been under-treated says the drug is theirs. Saying nothing because the arms are unknown puts the blind above the person, which is the inversion the whole trial exists to avoid.

ASK 9W

What should the nine participants be told this week?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

participants are told about problems that affect their own treatment · allocation
concealment: protecting the moment of entry — taken as read

VERDICT 92W

Two duties pull in different directions and both can be met. People are entitled to know about a fault that touched their own treatment, and telling them costs the trial nothing if the wording covers all nine equally. Saying they may have been under-treated says the drug is in the box. Saying nothing because the arms are unknown treats the blind as more important than the person, which is the inversion the whole trial is arranged to avoid. Waiting for the board is 12 more days for information that is already certain.

TAKEAWAY 18W

A participant is owed what affects them, said in a way that does not tell them their arm.

8.3 What the excursion record can actually prove

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · ATTEST

SCENE 36W

Nkechi Umeh has the site 19 excursion form, the logger trace, dispensing records and an inventory statement. Four claims on the form are signed. She has time to verify only two before the deviation file closes.

GUIDE 59W

Four signed claims on the excursion form, and time to verify two. Open each and read what backs it: the logger trace, the dispensing records, the inventory statement, or the form itself. Hold what the evidence does not support, and verify where a wrong claim would most damage somebody's later judgement about whether the drug was fit to use.

ASK 14W

Which claims most need verification before this record can support somebody else's judgement later?

BEHIND THE BUTTON 146W

What an excursion record is for. It documents a period when storage went outside its specified range, so that somebody later can decide whether the product was affected. It is written to be read by a person who was not there, which is exactly why the backing matters more than the wording.

Why the logger trace is not automatically the strongest evidence. It records what the logger saw at its position. Whether that position represents the product, whether the logger was calibrated, and whether the affected stock was in that location at that time are separate claims — and they are the ones usually signed rather than checked.

Why two checks and not four. The deviation file closes, and after that the record is what a reviewer sees. A claim held as unverified is honest; a claim verified after the file closes is a second deviation.

TAKES AS READ

a deviation record is evidence for later review · a signed statement records what someone asserts, not necessarily what the underlying records show

VERDICT 84W

A deviation form does not become reliable because somebody signed it. The duration can be checked against the logger, and the nine recipients can be checked against dispensing records. The important weak link is scope. If more than 11 kits were in the fridge, the size of the problem changes immediately. That claim needs an inventory reconciliation, not another signature. Recording the excursion still matters even if the final effect is small, because later reviewers can only challenge a judgement when its evidence survives.

TAKEAWAY 18W

A signed record is useful only when the claims that could change the conclusion are tied to evidence.

8.4 The clock that is not the analysis clock

SAFETY MONITORING · MONITORING BOARD ROOM · A ROOM · CALLBACK · TRIAGE

SCENE 35W

The second admission arrived 18 minutes ago. 1 patient is still on the ward, and the 2 sites are 400 kilometres apart. Vogt is reachable this afternoon. The next board meeting is 19 days away.

GUIDE 57W

Four options, and two clocks are running. One patient is still on the ward and the board meets in nineteen days. Ask of each option which clock it obeys. A serious unexpected event may change treatment tomorrow. Whether it also counts as an endpoint is a separate path, and the same event can travel both at once.

ASK 8W

What should happen to these two reports today?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

serious harm during a trial is reported whether or not the drug is thought to have caused it · hard endpoints against judged ones, and blinded adjudication — taken as read

VERDICT 90W

Safety reporting and endpoint analysis run on different clocks. A serious unexpected event may matter to treatment tomorrow, so it cannot wait 19 days for a board meeting. The site reports what happened now, with enough detail to judge urgency. Whether the event later counts toward a trial endpoint is a separate process. The same event can therefore travel down two paths at once: immediate safety reporting and later blinded adjudication. Waiting for one path to finish before starting the other loses time the safety system was built to protect.

TAKEAWAY 13W

A safety report does not wait for the analysis it might belong to.

END OF THE DAY 17W

A material problem becomes an analysis problem the moment it is not spread evenly across the arms.

What the guessing survey found

BEFORE THE DAY — CANVASS

Whether the sites are actually using the new consent form

The amendment went out three weeks ago and the analysis assumes it is in use. Ask round the sites until enough answers have come back to say whether they switched, because a trial half-consented under two versions has a problem nobody can fix retrospectively.

PLAN CARD 106W

Thursday, 11 days out, and a routine questionnaire has come back wrong. At the last annual visit every investigator was asked to guess which arm each of their participants was on. Across 1,180 guesses they were right 71% of the time. Ansah's committee, who see no arms at all, are unaffected; the symptom score written by the treating doctor is another matter, and it is the score Iriarte and Doyle have both called the one patients care about. Today you work out what a guessing rate like that does to each of CLARION-3's outcomes, and what the board has to be told before it reads either.

BRIEFING 13W

Investigators were asked to guess. They were right far more often than chance.

WHAT YOU DECIDE 14W

Find out whether the people scoring the outcome knew which arm they were scoring.

9.1 The blind that had already gone

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · CHOICE

SCENE 34W

At the annual visit, investigators and participants guessed each person's arm. Investigators were correct 71% across 1,180 guesses. The case packet also contains first-fortnight symptom logs, the allocation audit and the adjudicated outcome summary.

GUIDE 56W

Four explanations for guessing right 71 per cent of the time. Ask of each whether it accounts for both investigators and participants guessing well. A leaked list would not. A treatment effect big enough to be visible in improvement is not in the event data. And chance across 1,180 guesses does not reach anywhere near 71.

ASK 13W

What best explains investigators guessing the treatment arm correctly 71% of the time?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

blinding is meant to keep the person measuring from knowing the arm

VERDICT 74W

71% correct across 1,180 guesses is far from the 50% expected by chance. The visible flush also supplies a direct mechanism. Investigators can see it before they later score symptoms. A leaked allocation list could also raise guessing, but it would not explain participants guessing well too. Guessing from improvement would require a treatment effect much larger than the event data show. The survey therefore measures a real loss of blinding, not random noise.

TAKEAWAY 15W

When a treatment announces itself, the people measuring stop being blind whatever the protocol says.

9.2 Which measurements share the leak

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A PERSON · TRACE

SCENE 36W

Feldman has 4 CLARION-3 measurements on the wall. The guessing survey says the blind has leaked. Each measurement came through a different path, and the sponsor wants to know which evidence still stands on its own.

GUIDE 52W

Open each of the four measurements to see what it came through. Keep the ones that would still stand if the blind has leaked, and untick the rest. Then name the source the compromised ones share. A leak does not spoil everything equally, and the sponsor needs to know exactly what survives.

ASK 13W

Which measurements remain independent of the leak, and what source compromises the others?

BEHIND THE BUTTON 132W

What a blind is protecting. If people know which arm a patient is in, that knowledge can reach the measurement: through how symptoms are elicited, how ambiguous events are coded, how closely anybody is followed up. Blinding is what stops the treatment assignment influencing the record of what happened.

Why the guessing survey matters. Asking staff to guess the assignment measures whether the blind held. Guesses much better than chance mean it did not, which puts every measurement that passed through a person who knew under suspicion — and leaves the automated ones alone.

Why withdrawing everything is the wrong response. It discards measurements the leak could not have touched, and it tells a data monitoring committee that nothing is known. Naming what is compromised is what lets the trial continue honestly.

TAKES AS READ

the visible flush can reveal treatment assignment · adjudication removes the arm and site before the committee rules

VERDICT 82W

A blinding failure does not contaminate every number automatically. Bias needs a path. The treating clinician can see the flush, infer the arm and then write the symptom score. Participants can see the same flush and change how they take or return kit. The adjudication committee sees neither arm nor site, so that route is cut. The national death register is farther away still. Dependency tracing shows why four results pointing the same way are not automatically four independent pieces of evidence.

TAKEAWAY 19W

Agreement counts as independent evidence only when the measurements do not share the same route for bias to enter.

9.3 Telling the board what it does not yet know

SAFETY MONITORING · MONITORING BOARD ROOM · A ROOM · PROTOCOL

SCENE 36W

Vogt reads the interim pack in 11 days, and the PDF is queued for sign-off tonight. The guessing survey is absent because nobody expected it to matter. Balogun wants to leave it for the annual report.

GUIDE 64W

Four things about the trial's condition, and four ways a pack can carry them. Match each to the treatment that lets the board weigh it — some belong in the results section, one belongs in a limitations paragraph the board reads before the numbers, and one is genuinely an annual-report item. What decides the match is whether the board's next decision turns on it.

ASK 15W

Match each fact about the trial's condition to the way tonight's pack should carry it.

BEHIND THE BUTTON 146W

What the board is for. It is the only body that sees the arms separated, and it decides whether the trial continues. Everything it decides rests on the pack in front of it, so a weakness the pack does not mention is a weakness that cannot enter the decision.

Why a limitation is not an admission. A blinding survey suggesting the assignment may be guessable changes how much weight the effect estimate carries. Reported beside the estimate, it is information. Held for the annual report, it becomes something the board was not told before it decided, which is a different document to have to write.

What genuinely can wait. Enrolment running two months behind is a management problem with no bearing on whether the accumulated data support continuing. The test is not importance in general — it is whether this board's next decision turns on it.

TAKES AS READ

the board decides whether the trial continues

VERDICT 182W

A monitoring board's decision is made out of the pack, so the placement of a fact is not presentation — it decides whether the fact enters the decision at all. The blinding survey qualifies every efficacy number in the document, which is why it goes before them rather than after: a board that reads an estimate and then learns the blind may have leaked has already formed a view. Deviation rates belong site by site in the body, because a comparison is only as good as the sites feeding it and an average hides the two that are the problem. The event count is not a footnote either — 246 of 380 is the information fraction, and it fixes how much any estimate in the pack can settle. Recruitment running behind is real and belongs in the annexe, because nothing the board decides tonight turns on it. What makes the survey the

hard case is that leaving it out is defensible one item at a time and indefensible as a decision: the board cannot weigh a weakness it has not been shown.

TAKEAWAY 13W

A board cannot weigh a result whose weakness it has not been shown.

9.4 What counts as a deviation

SITE OPERATIONS · MONITORS' ROOM · A ROOM · CALLBACK · CHOICE

SCENE 32W

Lucia Marchetti has four things sites did this month that were not what the protocol says. None of them harmed anybody. Site 12 has asked which of them are worth the paperwork.

GUIDE 62W

All four are real deviations and none harmed anybody. Ask of each whether it lands on both arms equally. A late blood test and a late form are untidy in both arms. A different pump delivers the same drug. Only one of these changes who is in the trial. It matters because those entries need not be spread evenly across the arms.

ASK 10W

Which of these most threatens the comparison CLARION-3 is making?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the protocol states what should happen at each visit

VERDICT 86W

Deviations matter in proportion to what they do to the two arms. A late blood test and a late form are untidy and affect both arms equally, so the comparison survives them. A different pump delivers the same drug. Entering people the trial was not designed for changes who is in it — and if that site's entries are not spread evenly across the arms, it changes what is being compared. The test is always whether the two groups are still alike apart from the treatment.

TAKEAWAY 17W

A deviation is recorded for what it does to the comparison, not for how bad it looks.

END OF THE DAY 13W

A blind that has already failed cannot be restored by insisting on it.

What survives

PLAN CARD 110W

Friday, ten days out. Ansah's committee has re-read every ambiguous file from the past year with the site and the arm stripped, as always, and Petrescu has run the adjudicated endpoint for the closed report. The effect on the symptom score was large; on the adjudicated events it is smaller and it is still there. Feldman says that is the number the trial always had. Balogun says a smaller real effect is still a reason to stop giving people the other arm. Today you work out what the surviving result actually says, how precisely it says it, and what CLARION-3 does about the blind for the participants still being recruited.

BRIEFING 10W

Take the compromised score out and see what is left.

WHAT YOU DECIDE 11W

Re-read the trial on the outcome the unblinding could not reach.

10.1 Reading the same files twice

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · CLOUD

SCENE 30W

The committee has re-read 400 files. Ansah is asked whether a second reading by the same three people, who now know a guessing survey exists, can be trusted at all.

GUIDE 63W

The cloud is the 400 files, each plotted as the difference between its first and second reading. Place the middle bar and the two sigma bars to report where the differences centre and how widely they scatter, then buy what the committee can actually do to tighten the agreement. The band drawn across the plot is how much drift the endpoint definition tolerates.

ASK 20W

Report where the first-to-second-reading differences centre and how far they scatter, and get the spread inside what the definition tolerates.

BEHIND THE BUTTON 153W

What a difference of zero would mean. That the rule decided the reading, not the reader. A centre away from zero is drift — the second pass systematically scoring differently from the first — and a wide scatter is inconsistency, which is a different defect with a different fix.

Why the two are not the same problem. Re-centring a drifted committee is a recalibration: restate the rule, define the ambiguous case, and the middle moves. Only a sharper rule narrows the spread, and no amount of re-centring does it, which is what this panel is really about.

Why knowing the survey exists is dangerous. Three people who now expect the blind may have leaked are three people with a reason to read the ambiguous files differently, and their agreement with each other would not show it. Consistency between two readings comes from the rule, not from memory of what they decided last time.

TAKES AS READ

the adjudication committee never sees the arm · hard endpoints against judged ones, and blinded adjudication — taken as read

VERDICT 164W

A re-read is worth something only if the rule did the deciding. Plotted as first reading minus second, 400 files scatter around a centre near 0.6 events per hundred, with a spread of about 2.9. That is a committee both slightly drifted and quite inconsistent. Those are two defects, and they have different repairs. Restating the definition moves the centre and leaves the width alone. Defining the one ambiguous case narrows it. A fourth reader who has never seen these files narrows it again. Apply every re-centring and no rule change, and the spread sits exactly where it was. That is the lesson. Agreement between two readings by the same people can be manufactured by memory. Only a rule that decides the ambiguous case the same way

twice makes it evidence. The guessing survey matters for that reason. Three readers who suspect the blind has leaked have a reason to read the ambiguous files differently, and agreeing with each other would never reveal it.

TAKEAWAY 11W

Consistency between two readings comes from the rule, not from memory.

10.2 How big, and how wide

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A PERSON · BALLPARK

SCENE 39W

The press office has lifted “4.2 percentage points” into a draft briefing. Feldman circles the interval beside it. Before the number leaves the room, she wants the team to put both clinical size and sampling noise on the result.

GUIDE 60W

Five numbers, and two of them are answers to other questions: the risk ratio, and the risk difference already worked out. Ask of each what this calculation needs. The number needed to treat is built from the absolute difference, not the ratio. And precision is a separate question from size, which is why the interval is circled beside the headline.

ASK 14W

Estimate the NNT, then estimate the standard error of the 18.9% control-arm event rate.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the 2,400 participants are allocated about equally between 2 arms · a proportion estimated from a sample has sampling error · effect measures: relative risk, absolute risk reduction, number needed to treat — taken as read

VERDICT 81W

The risk difference and NNT describe the size of the observed benefit. Precision is a different question. For a proportion, standard error depends on both the event rate and the number measured. With about 1,200 people in the control arm, an 18.9% rate has an SE near 1.1 percentage points. A 95% margin is about twice that.

Increasing the sample narrows this uncertainty only with the square root of n , so halving the SE takes about four times as many people.

TAKEAWAY 20W

Effect size says how large the difference is; standard error says how much the estimate would move across repeated samples.

10.3 What to do about a blind that leaks

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A ROOM · CHOICE

SCENE 30W

600 participants are still in follow-up and the flush has not stopped happening. Diouf has been asked whether there is anything the pharmacy can do about it at this stage.

GUIDE 65W

Four options for a blind that has already leaked. Ask of each what it could still protect. Nothing recovers the blind for anybody who saw the first fortnight. What can be protected is the route from belief into the number. Matching the comparator is the right design, four years too late. And a limitation paragraph fixes nothing while six hundred people are still being scored.

ASK 10W

What is the best available fix for the remaining follow-up?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the drug produces an early visible effect that the comparator does not

VERDICT 95W

The blind has already failed for anybody who saw the first fortnight, and nothing recovers it for them. What can still be protected is the route from belief into the number: separate the assessor from the treating team, and the score is written by somebody with nothing to infer from. Matching the comparator's appearance is the right design and it is four years too late to introduce here. Leaving it in the notes matters — the notes are the source record — and a limitation paragraph fixes nothing while 600 people are still being scored.

TAKEAWAY 15W

When a treatment cannot be hidden, protect the measurement instead of pretending the blind holds.

10.4 Ten minutes on each case

SITE OPERATIONS · MONITORS' ROOM · A ROOM · CALLBACK · VALUE

SCENE 38W

Lucia Marchetti has ten minutes with each site before the reports go to Vogt. One coordinator is at a bedside. Five useful questions are on her screen, but asking all of them would take longer than either call.

GUIDE 59W

Ten minutes with each site, five useful questions, and no time for all of them. Open each question and ask what its answer would change before the reports go to Vogt. One coordinator is at a bedside, so the call has to be short and worth having. Spend the ten minutes on what the reports cannot already tell you.

ASK 13W

What should Marchetti spend the ten minutes finding out before the reports leave?

BEHIND THE BUTTON 132W

Why some questions are wasted on a call. Anything the site has already entered in the system is in the reports. A call is the only way to get at what has not been written down: why a visit slipped, what the coordinator is worried about, whether a protocol step is being interpreted differently.

Why 'before the reports leave' is the constraint. Once the report is out, a correction is a deviation and a conversation. Something learned now changes what the report says; the same thing learned tomorrow changes what has to be explained.

What a coordinator at a bedside can actually do. Answer one or two things from memory. A call that asks five questions gets three rushed answers, and the rushed ones are the least reliable data in the trial.

TAKES AS READ

a safety pattern depends on whether cases share timing and clinical features · a VALUE board charges time for each piece of evidence · eligibility criteria, and what widening them changes — taken as read

VERDICT 79W

Two reports become a signal only when their details line up in a way that could matter. Timing is the fastest discriminator here. A case three days after first exposure tells a different story from one eight months later. Matching intervals in two distant sites would strengthen the case for a shared cause. Different intervals would weaken it. Investigator opinions, recruitment totals and withdrawal status may still matter, but none can establish that temporal pattern before the reports leave.

TAKEAWAY 19W

When time is scarce, collect the evidence that can change the decision before collecting detail that only decorates it.

A finding that shrinks when the bias is removed was partly the bias.

The quiet day

PLAN CARD 108W

Monday, six days out, and nothing is wrong. Queries are down to 41, the sites have sent what they owe, and Renner's reconciliation balanced on the first run for the first time in 2 years. Umeh has the lock plan on the wall with Thursday circled. Vogt reads the pack on the following Tuesday and Petrescu is the only person who will have seen it before then. Today you close the file: decide what still has to be true before the data can be frozen, what a locked database actually promises, and which of the 41 open queries has to be answered before Thursday rather than after it.

BRIEFING 9W

Nothing has gone wrong. The cut is on Thursday.

WHAT YOU DECIDE 8W

Close the file that the board will read.

11.1 Freezing something that was moving

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · CHOICE

SCENE 37W

The database-lock checklist is projected on the wall and the cut is three days away. Umeh has one question from the sponsor and another from Renner: once the file freezes, what exactly would justify opening it again?

GUIDE 63W

Four things a lock might buy. Ask of each whether a frozen file could deliver it. It cannot make the data correct, and it cannot stop collection. What it does is put a visible boundary around what the analysis used, so anybody can rerun it. Reopening is possible and creates a new version of the evidence, which is why it needs a reason.

ASK 9W

What does locking the database buy the interim analysis?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

data arrive from sites continuously while a trial runs

VERDICT 90W

A database lock creates a version of the trial that cannot quietly move under the analysis. Queries are closed, coding is fixed and the dataset is frozen before the planned calculation runs. That makes the result reproducible later. Reopening is possible, but it must be exceptional and documented because it creates a new version of the evidence. A discovered error that changes a primary event can justify that step. A cosmetic correction usually cannot. The point is not permanence; it is a visible boundary around what the analysis actually used.

TAKEAWAY 15W

Locking fixes what the analysis was run on, so the result can be checked later.

11.2 3 days, 41 queries

SITE OPERATIONS • MONITORS' ROOM • A PERSON • DELEGATE

SCENE 39W

41 queries remain and the interim file locks in 3 working days. 1 concerns a primary-event date, 1 a duplicated visit, and 39 are contact-field corrections. Renner has two people available, and you can keep only one problem yourself.

GUIDE 49W

Forty-one queries, three working days, two people, and one problem you keep yourself. For each thing you hand over, name the owner, the first action, and what brings it back to you. Thirty-nine of the queries are contact-field corrections, which is a volume problem. The other two are not.

ASK 17W

What do you keep, what do you hand off, and what must each delegated owner bring back?

BEHIND THE BUTTON 133W

Why volume and judgement are different work. Thirty-nine identical corrections need somebody methodical with a clear rule for the ambiguous ones. A primary-event date and a duplicated visit each need somebody who understands what the field means, because getting them wrong changes the analysis rather than the tidiness of the file.

Which to keep. The primary-event date is the one that feeds the interim analysis directly. A wrong date can move an event across the analysis boundary, which is the one error in this list that changes a result rather than a record.

What a return condition looks like here. Not 'come back if stuck'. A rule: any correction that would change a date, or any duplicate that cannot be resolved from the source document, comes back to you rather than being decided.

TAKES AS READ

an event outside the cut is absent from this interim analysis • delegation needs an owner, a first action and a condition for returning the problem

VERDICT 85W

The biggest pile is not the most urgent problem. A missing event date can decide whether a primary event exists in Thursday's analysis at all. Once the file is cut, that omission cannot be repaired for this interim. The duplicate visit is important but stable, and a site monitor can reconcile it against the source note. Contact-field corrections are numerous and visible, yet they do not change the endpoint count. Good delegation protects the irreversible decision while giving stable work clear owners and return rules.

TAKEAWAY 18W

Urgency belongs to the problem whose consequence becomes irreversible first, not to the problem making the most noise.

11.3 Nothing left in the tray

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · CHOICE

SCENE 32W

Ansah has 19 files still under adjudication and 3 days. 2 of the 19 are from participants recruited in the first year and have been waiting on 1 missing discharge summary each.

GUIDE 59W

Four reasons the tray has to be empty. Ask of each what an unadjudicated file actually is in the analysis. Adjudication is what turns something that happened into a countable event, so nineteen files in a tray are nineteen events nobody had. And files wait longest where the evidence is thinnest, which is not evenly spread across the arms.

ASK 12W

Why must the committee clear its tray before the file is locked?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an event is only counted once the committee has ruled on it

VERDICT 85W

Adjudication is not paperwork after the fact; it is the step that turns something that happened into a countable event. 19 files sitting in a tray are 19 events the analysis will behave as though nobody had. That is not neutral: files wait longest where the evidence is thinnest, and the thin files are not evenly spread between arms. Rulings made after the lock go into the next version, which means the interim was read on a trial that was quietly missing part of itself.

TAKEAWAY 12W

An unadjudicated event is invisible to the analysis, whatever the notes say.

11.4 What an extra look spends

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · CALLBACK · BALLPARK

SCENE 38W

The medical director wants one unplanned endpoint summary before noon and has already drafted the email asking for it. Feldman has the two planned looks on the whiteboard and an empty box where this new request would sit.

GUIDE 61W

Add up what the planned looks already spend, and compare it with what the trial has to spend in total. Each look has a boundary, and the error rate it uses is the price of that look. Pick the three planned amounts. The unplanned look's price is on the board too, and it is what the sum has no room for.

ASK 12W

What do the planned looks already spend of the trial's error budget?

BEHIND THE BUTTON 167W

What is being spent. A five per cent error rate is a promise about the whole trial, not about any one test. Every look at the accumulated data is another chance to catch a wandering difference at its most flattering moment, so each look consumes part of that budget and the budget does not refill.

How planned looks are paid for. The boundaries at the planned looks are made harder — an interim result has to be more extreme than the final one to stop the trial — and the amounts add up to no more than the total. That is the whole of the arithmetic: the spending plan is written before any data arrive.

Why an unplanned look is different in kind. It has no boundary and no allowance, so whatever it spends comes out of the final analysis. A trial that has spent more than its budget can no longer say what its error rate is, which is a worse position than not having looked.

TAKES AS READ

a trial accepts a small chance of calling a difference real when there is none · type I and type II error, and what α and β are for — taken as read

VERDICT 156W

The rule is $\Sigma \alpha_{\text{spent}} \leq \alpha$. This trial is run one-sided at 0.025, and the spending plan puts 0.001 at the first interim, 0.004 at the second and 0.020 at the final analysis: $0.001 + 0.004 + 0.020 = 0.025$, which is the whole budget and exactly what the plan is allowed to be. That is why the interim boundaries are so hard to cross — an early look is cheap precisely because it demands an extreme result. An unplanned look priced at 0.015 takes the total to 0.040, and the trial can no longer state its own error rate: 4 per cent of trials with no real effect would report one. Nothing about the data changes when somebody looks, which is what makes this counter-intuitive. What changes is the number of chances the trial has taken to be fooled, and that number is in the denominator of every claim it goes on to make.

TAKEAWAY 14W

A trial's tolerance for error is spent by testing, and it does not refill.

END OF THE DAY 12W

A locked file is a promise about a moment, not about perfection.

Three things at once

BEFORE THE DAY — EVADE

The sponsor's medical director wants a number before the board sees it

He has been outside statistics since eight wanting the interim effect size, and he is not on the list of people who may have it. Keep clear of him until the board has read the pack. A number that leaks before the committee sees it cannot be un-leaked.

PLAN CARD 100W

Tuesday, five days out, and three things landed before nine. Petrescu's closed tables show 1 subgroup — participants over 70 — crossing the boundary on its own. A group in Lyon has published a smaller trial of a related drug with a strong result. And the regulator has asked for the liver cases from day three as a formal safety update, due Friday. Balogun has read the Lyon paper twice and wants the board told about all three together. Today you decide what goes in the pack, in what order, and what each of the three is actually evidence about.

BRIEFING 15W

A subgroup has crossed its line, a rival has published, and the regulator has written.

WHAT YOU DECIDE 13W

Rank what the board is told when everything arrives on the same morning.

12.1 The subgroup that crossed

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · BALLPARK

SCENE 32W

Participants over 70 show a difference that clears the line on its own. Feldman asks which of 14 prespecified subgroups this is, and whether anybody has looked at the other 13 yet.

GUIDE 57W

Put an interval around the over-70 result before deciding what it is worth. The interval is the estimate plus and minus about two standard errors, so pick the multiplier and the standard error and work out the half-width. One tile is the standard error of the whole trial, which is the number this subgroup does not have.

ASK 14W

How wide is the interval around the over-70 result, either side of the estimate?

BEHIND THE BUTTON 152W

What the width says. A confidence interval is the estimate give or take about two standard errors, and the standard error grows as the group shrinks. A subgroup of a few hundred therefore carries an interval several times wider than the trial's, which is why subgroup estimates swing so far from the overall result.

Why 14 subgroups is the other half of the problem. Testing fourteen slices at the usual threshold gives about a one-in-two chance that at least one clears the line by luck alone. A subgroup that crossed is therefore what an honest trial expects to see, not evidence that something is different about people over 70.

What the over-70 result is for. It is a hypothesis somebody could test in a trial designed to test it. Stated as a finding it becomes a treatment recommendation for a group whose interval, once you draw it, includes no benefit at all.

TAKES AS READ

a subgroup is a slice of the participants analysed on its own

VERDICT 164W

A confidence interval is $CI \approx \text{estimate} \pm 1.96 \times SE$. The over-70 subgroup's absolute risk reduction is 6.0 percentage points with a standard error of 3.4, so the half-width is $1.96 \times 3.4 \approx 6.7$ and the interval runs from about -0.7 to 12.7 points. It includes zero, which is the whole answer: the subgroup that 'crossed its line' has an interval that does not exclude no effect once it is drawn properly. The trial as a whole has a standard error of 1.5, so its interval is less than half as wide — precision improves with the square root of the sample, and a subgroup is a smaller sample by construction. Then there is the second problem: with fourteen prespecified subgroups, the chance that at least one clears the usual threshold by luck alone is about one in two. So this result is what a well-run trial expects to produce, and the honest sentence calls it a hypothesis rather than a finding.

TAKEAWAY 18W

A subgroup result is one of many results, and it has to be judged as one of many.

12.2 The update that is due Friday

SAFETY MONITORING · MONITORING BOARD ROOM · A ROOM · CHOICE

SCENE 29W

The regulator wants the liver cases written up by Friday. There are now four, not two. Vogt has seen all four by arm; nobody else in the building has.

GUIDE 62W

Four versions of Friday's update. Ask of each whether a reader could work out how common the problem is. Four cases among forty exposed people is a different thing from four among twelve hundred. So a numerator without a denominator is not a rate. Causality opinions can go in and do not replace the counts, and efficacy belongs to a closed review.

ASK 9W

What must Friday's update contain to be worth reading?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

serious events are reported to regulators on a stated schedule

VERDICT 79W

Four cases do not say how common the problem is. 4 cases among 40 exposed people mean something very different from 4 among 1200. The regulator needs both the numerator and denominator, plus enough clinical detail to recognise a pattern. Causality opinions can be included, but they do not replace the counts. Efficacy results belong to the board's closed review. A recommendation about continuing also comes later, after the people responsible for that decision have the same safety evidence.

TAKEAWAY 14W

A safety update reports the events and the denominator, not a conclusion about them.

12.3 What the Lyon paper is worth tonight

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · VALUE

SCENE 36W

The Lyon trial published a large benefit at 6:10 this morning. By lunch, Balogun wants CLARION-3 to respond. Nkechi Umeh has the paper, its supplement and four review hours before she must state the trial's position.

GUIDE 51W

Four review hours before a position has to be stated, and a paper published this morning. Open each piece of the Lyon evidence and ask what would change about CLARION-3's position depending on how it comes out. Buy what could change the position, not what would explain the headline more thoroughly.

ASK 14W

Which pieces of the Lyon evidence are worth buying before tonight's position is written?

BEHIND THE BUTTON 132W

Why the headline is rarely the useful part. A large benefit in another trial matters to this one only through specifics: the population, the endpoint definition, the comparator, whether the effect is plausible in the patients CLARION-3 enrolled. Those live in the supplement rather than the abstract.

What a position actually has to survive. Balogun will be asked whether CLARION-3 should continue, and the answer has to hold whether the Lyon result replicates or not. Evidence that would only be reassuring if Lyon is right is not worth an hour.

Why four hours is the whole point. There is time to read some of it properly or all of it badly. Choosing which parts is the same judgement every VALUE stop asks for, on the day it is least comfortable to make.

TAKES AS READ

a related trial can change the uncertainty that makes continued randomisation defensible · an outside result does not rewrite a prespecified analysis plan

VERDICT 82W

A striking outside result is neither ignored nor copied into the current trial. First ask whether it measured the same kind of outcome, on enough events, with uncertainty narrow enough to matter. Then ask whether its randomisation protected the comparison. Those features determine how much the paper should change the board's uncertainty. A plausible mechanism or an impressive subgroup can support interpretation, but they cannot rescue an incomparable endpoint. The result belongs in the board's discussion, while CLARION-3's registered analysis remains fixed.

TAKEAWAY 19W

External evidence matters in proportion to how much it can change the decision, not how dramatic its headline sounds.

12.4 What has to go out today

SAFETY MONITORING · MONITORING BOARD ROOM · A PERSON · SPOT

SCENE 48W

Adverse event reports queue on the safety desk faster than they can all be assessed. The rule about which go out today is set by the safety physician, and it changes as the picture does — after the second case, after the regulator writes, after the board meets.

GUIDE 52W

Send out what the current rule requires and hold the rest for assessment. The rule on the desk changes during the day and nobody announces it. What is scored is the reports either side of a change, because they are the only ones that show whether the desk is reading the rule.

ASK 11W

Report to the rule in force, and keep watching the rule.

BEHIND THE BUTTON 104W

Why the rule moves at all. Ordinarily what goes out today is anything serious and unexpected. Once a second case of the same thing appears, the rule becomes anything resembling it, whether or not it is serious. After a regulator asks, it is anything they have asked about, on their timetable rather than the trial's.

Why late is not a small failure. Expedited reporting exists so that somebody outside the trial can see a pattern the trial cannot. A report held for assessment is a report that is not in that view, and assessment is exactly what the recipient is supposed to do independently.

TAKES AS READ

a trial reports certain events without waiting to decide what caused them

VERDICT 137W

Three rules run across the day and each wants a different part of the queue. Anything serious and unexpected, then anything resembling the cluster, then anything the regulator has asked about. A report can answer two at once, which is what makes the change cost real rather than notional. The panel scores the window either side of each change, because most of the queue is wanted by neither rule and is correctly held for assessment by somebody who has read nothing. The desk's own version is the hour after a second case appears: the rule has changed to anything resembling it, and the reports going out are still the ones that were serious-and-unexpected when the morning started — which is precisely the hour in which the pattern was visible and nobody outside the trial could see it.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

12.5 When the blind is opened for one person

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A ROOM ·
CALLBACK · SEQUENCE

SCENE 37W

Diouf has a call from an intensive care registrar at three in the morning, on speaker. A CLARION-3 participant is in front of him and the treatment he is considering depends on what she has been receiving.

GUIDE 55W

Four steps, and the rail is not a clock — every one of them could be done in the next ten minutes. Order them by what each one protects: the thing that cannot be recovered if it is skipped goes first, and the thing that can be done afterwards without costing anybody anything goes last.

ASK 13W

Order the four steps by what is lost if the step is skipped.

BEHIND THE BUTTON 170W

Why the blind exists. It protects the comparison, not the participant. Every person who knows an assignment is a person whose judgement can drift with it — the treating team, the assessor, the participant. The trial's answer depends on that ignorance being intact across four hundred people, and one code being opened costs one of them.

Why the clinical decision comes first anyway. The blind is a research instrument and this is a patient in front of a registrar at three in the morning. If knowing the assignment changes what is safe to give her, the code is opened; the trial can lose a participant from the blinded analysis and cannot lose her.

What has to be written down and why the order matters. The reason, the time and who authorised it, recorded before the envelope is opened rather than reconstructed afterwards — because once the assignment is known, nobody can prove the request was made on clinical grounds. Telling the monitoring committee can wait; telling the site file cannot.

TAKES AS READ

each participant's assignment is held in a sealed record that can be opened · blinding, and the several people it can fail for separately — taken as read

VERDICT 139W

The first step cannot be recovered, because it cannot be established after the fact. Once the registrar knows the assignment, nobody can show the request rested on clinical need rather than curiosity. The written reason is the same problem in another form. A note made before the envelope is opened is evidence. One made afterwards is an account. Only then is the code opened, and the participant treated on what is known. Notifying the committee and flagging her in the analysis plan matters, and costs nothing by waiting until morning. That is why it is last rather than unimportant. The blind protects the comparison across four hundred people. It

is opened for one of them when a decision for that person turns on it. The price is paid in the analysis, not argued about at three in the morning.

TAKEAWAY 17W

The blind exists for the trial; it is opened when a decision for one person needs it.

END OF THE DAY 15W

Three urgent things are not one big thing, and the board is owed the difference.

1 of 14

PLAN CARD 97W

Wednesday, four days out. The prespecified subgroup list has 14 entries on it, and the over-seventies are the eleventh. Feldman has spent the morning showing what 14 looks like: with that many slices, one of them clearing the usual line is close to expected even when the treatment does the same thing in everybody. Balogun, who has argued all fortnight for acting sooner, agrees before she has finished. Today you put a number on how often 14 independent looks throw up a false positive, and decide what the board is told about the over-seventies as a result.

BRIEFING 10W

Feldman has the list of every subgroup the plan named.

WHAT YOU DECIDE 14W

Work out what the over-seventies result would have cost if it had been believed.

13.1 How often something clears the line by luck

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · BALLPARK

SCENE 34W

14 prespecified subgroups, each tested at the usual 5%. Feldman writes the arithmetic on the board and asks what the chance is that at least one clears the line with no real difference anywhere.

GUIDE 56W

Five numbers, and two of them belong elsewhere: the number randomised, and the one that stands for certainty. Ask of each whether this stacking depends on it. Each test carries its own small chance of misfiring, and those chances multiply rather than staying small. Real subgroups overlap, which softens the number without changing what it says.

ASK 16W

Estimate the chance that at least 1 of 14 subgroups clears the line by luck alone.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

each test has a 5% chance of a false positive on its own

VERDICT 87W

Each test carries its own small chance of being wrong, and the chances do not stay small when they are stacked. With independent tests the probability that none misfires is 0.95 multiplied by itself once per test, so 14 tests leave barely half the outcomes clean. That is the arithmetic behind every impressive subgroup: not that anybody cheated, but that a room looking in 14 places will usually find something in 1 of them. Real subgroups overlap and correlate, which softens the number without changing the lesson.

TAKEAWAY 15W

The chance of at least one false positive rises fast with the number of tests.

13.2 The slice, and what it was scored on

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A PERSON · CHOICE

SCENE 33W

The over-seventies result is on the symptom score rather than the adjudicated event. Ansah points out that this is the outcome the guessing survey landed on, and that older participants attend most often.

GUIDE 57W

Four accounts of what the subgroup result rests on. Ask of each whether the outcome named is the protected one or the leaky one. Two weaknesses compound here: one of fourteen looks, measured on the score the guessing survey reached. Older participants attend more often, so there are more scored visits for a small bias to enter.

ASK 7W

What does the over-seventies result rest on?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the symptom score is written by the treating doctor and the event is adjudicated

VERDICT 80W

Two weaknesses compound here. The finding is 1 of 14 subgroup looks, so chance already has many opportunities to produce something interesting. It is also measured on the symptom score, which the guessing survey showed can inherit treatment knowledge. Older participants attend more often, creating more scored visits where a small bias can enter. The adjudicated event in the same subgroup shows nothing unusual. That contrast matters: the subgroup signal is strongest exactly where the measurement process is least protected.

TAKEAWAY 13W

A subgroup finding inherits every weakness of the outcome it was measured with.

13.3 What to say about a slice

SAFETY MONITORING · MONITORING BOARD ROOM · A ROOM · CHOICE

SCENE 30W

Vogt will see the over-seventies table whatever the pack says about it. What the pack can do is tell her what she is looking at before she looks at it.

GUIDE 59W

Four ways to present one slice, and the board will see the table whatever the pack says. Ask of each what reaction the layout invites. A striking slice alone invites the reader to treat it as the finding. The eleventh row of fourteen invites them to count. And filtering what looks unlikely starts making the board's decision for it.

ASK 10W

How should the over-seventies table be presented to the board?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

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TAKES AS READ

the board sees the trial's subgroup tables in closed session

VERDICT 86W

A single striking slice on its own page invites the reader to treat it as the finding. The same slice as the eleventh row of 14 invites them to count, which is the right reaction. Presenting all of them costs nothing and removes the choice of what to emphasise from whoever assembles the pack. Leaving it out is worse than either: the board is the body entitled to see everything, and a pack that filters what looks unlikely has started making the board's decision for it.

TAKEAWAY 13W

A subgroup table is reported with the number of looks that produced it.

13.4 What has to be told, and how fast

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · CALLBACK · PROTOCOL

SCENE 29W

Umeh has a wall calendar with two windows marked on it in red and a stack of four events from this month. She has to sort them before Friday.

GUIDE 61W

Four events and four obligations. Pair them by asking three things of each: is it serious, was it unexpected, and could it be related? The combination sets the clock, not how dramatic it sounds. A death nobody thinks is related is still serious. And an admission for the trial's own condition is the thing being counted rather than a safety report.

ASK 10W

Match each event to what the trial owes and when.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

regulators set deadlines for reporting harm during a trial · hard endpoints against judged ones, and blinded adjudication — taken as read

VERDICT 86W

The reporting window is set by the category, not by the drama. Serious, unexpected and possibly related is the combination that runs on a clock of days, because it is the combination that might change how the drug is used tomorrow. A death nobody thinks is related is still serious and still reported, on the ordinary schedule. An admission for the trial's own condition is the thing being counted, so it goes to adjudication rather than to the safety desk. Everything else lives in the file.

TAKEAWAY 16W

Expedited reporting is decided by what the event is, not by what anybody thinks caused it.

END OF THE DAY 12W

A result that would be surprising once is unremarkable in 14 attempts.

The last reversible moment

PLAN CARD 108W

Thursday, one day out. The pack is printed, the file is locked, and Petrescu has run the analyses the plan named. Tomorrow Vogt's board sits for four hours and recommends one of three things: stop for benefit, stop for futility, or carry on to the end. Balogun will attend the open session and say what he has said all fortnight; Doyle will be in the room too, and has asked what a participant recruited last month is owed by either decision. Today you assemble what the board needs to weigh — how likely the trial is to finish positive, and what each way of stopping would leave unknown.

BRIEFING 10W

The board meets tomorrow. After that the decisions are made.

WHAT YOU DECIDE 15W

Prepare the case the board will decide on, and be honest about what stopping costs.

14.1 Three things the board can say

SAFETY MONITORING · MONITORING BOARD ROOM · A ROOM · PROTOCOL

SCENE 38W

Vogt's board meets behind a closed door in 40 minutes. The adjudicated effect is modest, the stopping boundary is not crossed, and four liver cases remain unexplained. Three recommendations are printed on separate cards in front of her.

GUIDE 57W

Three recommendations and four costs, and none of the choices is free. Pair them by asking what each one gives up: precision, a real benefit, another year of exposure, or comparability with the registered plan. Early stops overstate effects. And even changing what gets reported has a cost, unless the change is recorded before it is used.

ASK 12W

Match each recommendation to what the board gives up by making it.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a monitoring board recommends and the sponsor decides · sample size and statistical power — taken as read

VERDICT 85W

None of the board's three choices is free. Stopping for benefit before the boundary is crossed risks exaggerating an effect and loses future safety follow-up. Stopping for futility can discard a modest benefit that more events might confirm. Continuing preserves the planned evidence but keeps people exposed to the arm doing worse so far. Even changing what gets reported has a cost. The registered plan is part of why readers can trust the result, so any change has to be recorded before it is used.

TAKEAWAY 12W

Each way of ending a trial buys something and gives something up.

14.2 What the rest of the trial would probably do

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A PERSON · BALLPARK

SCENE 35W

Feldman has computed what the trial would most likely conclude if it ran to 380 events, given the 246 in hand. Balogun asks the obvious question: if it is going to end positive, why wait.

GUIDE 59W

The forecast rests on one number: how extreme the remaining events would have to be for the final analysis to clear its boundary. Work out the information fraction first — events in hand over events planned — and the verdict turns that into the conditional power. One tile is the events still to come, which is not the fraction.

ASK 16W

How far through its information is the trial, and what does that make of the forecast?

BEHIND THE BUTTON 158W

What the information fraction is. Events observed over events planned. It is how far through an event-driven trial actually is, and it is not the same as how far through its calendar or its recruitment — 246 of 380 events is about two thirds of the information, whatever the enrolment log says.

What conditional power is. The probability that the trial is significant at the end, given the data so far and assuming the effect really is what the protocol assumed. It is a forecast about a study: it answers "will this trial clear its boundary", and not "does this treatment work".

Why a high forecast is not a reason to stop. Stopping early converts a forecast into a conclusion and spends error rate that was budgeted for the final look. The trial was designed to answer the question at 380 events because that is how much information the question needs, and a probable answer is not one.

TAKES AS READ

the trial has collected about two thirds of its planned events · sample size and statistical power — taken as read

VERDICT 185W

The information fraction is $t = \text{events observed} / \text{events planned}$, so $246 \div 380 = 0.65$: about two thirds of the way through, whatever the calendar says. That fraction is what the forecast is built on. Conditional power = $P(\text{significant at the end} \mid \text{data so far})$, and with the interim z-score at 1.42, a final boundary of 1.96 and 35 per cent of the information still to come, the remaining events would have to be mildly favourable rather than dramatic — which comes out at roughly 0.82. Two things follow. It is a forecast about a study, not a statement about the treatment: it assumes the effect is what the protocol assumed and asks whether this trial will clear its line. And a probable answer is not an answer — stopping now converts the forecast into a conclusion and spends error rate that was budgeted for 380 events. The mirror case

is the one the format exists for: a conditional power of 0.05 is a futility signal, and it is still a statement about the trial rather than proof the treatment does nothing.

TAKEAWAY 12W

A forecast about a study is not a statement about the treatment.

14.3 What a stop obliges

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · SEQUENCE

SCENE 35W

Umeh has the registry entry, protocol and publication policy open before the board call. One participant is due an infusion on Monday. Doyle asks what happens, in what order, if the board recommends stopping tomorrow.

GUIDE 54W

All four steps will happen, so ask who learns what, and when. The board recommends and the sponsor decides. Somebody is due an infusion on Monday. A registered trial that quietly disappears is how the literature acquires missing arms. And a trial stopped early still owes its results, with the shortened follow-up stated plainly.

ASK 11W

Order what has to happen if the board recommends stopping tomorrow.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a trial's commitments continue after recruitment ends · allocation concealment: protecting the moment of entry — taken as read

VERDICT 84W

A stop is an event with a sequence, and the sequence is about who learns what and when. The sponsor decides, because the board only recommends. Sites and participants come next and quickly: somebody is due an infusion on Monday and is entitled to know. The public record is corrected because a registered trial that quietly disappears is how the literature acquires missing arms. And reporting is not optional — a trial stopped early still owes its results, with the shortened follow-up stated plainly.

TAKEAWAY 13W

Ending recruitment does not end the trial's obligations to the people in it.

14.4 What a short screening log says

SITE OPERATIONS · MONITORS' ROOM · A ROOM · CALLBACK · CHOICE

SCENE 30W

31 screening logs on 1 page. Most sites screen four and enter one. Site 12 screens 11 and enters 9, and its consent rate is the highest in the trial.

GUIDE 57W

Four explanations for a short screening log. Ask of each what the log would look like if it were true. The high entry rate is not the strongest clue. The nearly empty reasons column is. Eleven screened and nine entered should still record why two were not, and a year should show many who failed before consent.

ASK 8W

Which explanation best fits site 12's screening log?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

sites record everybody they assess as well as everybody they enter · eligibility criteria, and what widening them changes — taken as read

VERDICT 77W

The high entry rate is not the strongest clue. The nearly empty reasons column is. A site that truly screened 11 people and entered 9 should still record why the other 2 were not entered. Across a year, it should also show many people who failed before consent. If those people never reach the log, the trial cannot describe who was considered and passed over. The result is a selected group whose recruitment funnel cannot be reconstructed.

TAKEAWAY 14W

Recruitment figures describe a funnel, and an empty log hides the top of it.

END OF THE DAY 17W

Stopping is a decision about what will never be known, as much as about what is known.

What it is worth saying

BEFORE THE DAY — TAG

Catch Ansah before the adjudication committee sits

Ansah, the endpoint adjudication chair, has the four re-read files that changed classification, and once the committee is in session the door shuts for three hours. Whether those four moved is the difference between a signal and a rounding error in this afternoon's summary.

PLAN CARD 99W

Friday, and the board reported yesterday: continue to the planned events, with the symptom score dropped from the primary reporting and the liver cases followed formally. Balogun has accepted it. Feldman has written the sentence she thinks the trial can defend; the sponsor's communications team has written a different one, using the same numbers. Doyle wants to know which of the two would help somebody decide on a Tuesday afternoon. Today you settle how the result is stated, what the trial is entitled to claim about its endpoint, and what happens now to the 2400 people still in it.

BRIEFING 13W

The board has reported. Now the trial has to be described in words.

WHAT YOU DECIDE 11W

Say what CLARION-3 found, how strongly, and what nobody knows yet.

15.1 One relative risk, many absolute effects

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · DEGENERACY

SCENE 39W

2 lines for the board chair's closed briefing describe the same CLARION-3 result. 1 says risk fell about 22%; the other says roughly 4 events were prevented per hundred people. Umeh puts both arm rates on a two-control display.

GUIDE 56W

Two controls slide the two arm risks along a family of pairs that all give the same relative result. Find out how wide that family is before choosing a line for the briefing. Then bring in the second measurement, which fixes one arm rather than the ratio, and watch the family collapse to a single pair.

ASK 18W

How many pairs of arm risks fit the relative result, and what second measurement identifies the CLARION-3 pair?

BEHIND THE BUTTON 147W

Why a relative risk does not pin the arms. A 22% reduction is a ratio. Twenty per cent against sixteen has it; two per cent against one and a half has it too. The ratio is the same and the number of people helped differs by an order of magnitude.

What the absolute figure adds. Four events prevented per hundred people is a statement about how common the outcome is, which the ratio deliberately hides. Both sentences are true of the same result, and only one of them tells the chair what the trial would do for a patient.

Why boards are shown both. A relative figure travels well between populations and exaggerates importance in rare outcomes. An absolute figure is honest about scale and does not transfer to a population with a different baseline. Presenting one alone is the most common way a true result misleads.

TAKES AS READ

relative risk is treatment risk divided by control risk · absolute risk reduction is control risk minus treatment risk

VERDICT 79W

A ratio does not carry the baseline rate with it. Treatment and control risks of 3.9 and 5% give almost the same relative reduction as 14.7 and 18.9%. The absolute benefits are very different. That is why the relative result forms a whole line of possible pairs on the display. Adding the measured absolute reduction gives a second line. Their intersection recovers the actual arm rates and shows why an absolute statement carries information the relative headline leaves out.

TAKEAWAY 15W

A relative effect does not determine the absolute benefit until the baseline risk is known.

15.2 What can be claimed

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A PERSON · CHOICE

SCENE 30W

Ansah is asked to write one sentence about the adjudicated endpoint for the board's minutes. He has the effect, the interval, and the fact that the trial has not finished.

GUIDE 53W

Four sentences for the minutes. Ask of each what it claims beyond the estimate and its interval. One refuses to say anything until the target is reached, which throws away information already collected. One imports a fragile subgroup. One argues the trial should have stopped, which is the question the boundary already answered.

ASK 8W

Which sentence does CLARION-3's adjudicated endpoint currently support?

BEHIND THE BUTTON 195W

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What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the interval around an estimate says what the data are compatible with

VERDICT 81W

An interim estimate is evidence, but it is not the final word. The point estimate says where the observed difference sits, and the interval shows the range still compatible with the data. More events can move both. Saying there is no effect until the target event count is reached throws away information already collected. Claiming the effect is specific to older participants imports a fragile subgroup result. Claiming the trial should have stopped ignores the stopping boundary the board actually used.

TAKEAWAY 13W

An honest claim carries its uncertainty and the conditions it was measured under.

15.3 What happens to the people in it

SITE OPERATIONS · MONITORS' ROOM · A ROOM · CHOICE

SCENE 36W

The board has said continue. At 8 Monday morning, 31 sites will open their phones and 2,400 participants will still be enrolled. Marchetti has one letter to tell every site what changes, and what does not.

GUIDE 57W

Four things participants might be owed. Ask of each whether it could travel back into the trial through them. Consent is a continuing decision, not a signature from four years ago, so saying nothing fails an obligation. But releasing the interim numbers or somebody's own arm lets treatment knowledge spread back through the sites still scoring outcomes.

ASK 11W

What are the participants owed now that the board has reported?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

participants continue treatment and follow-up until the trial ends · allocation concealment: protecting the moment of entry — taken as read

VERDICT 87W

Consent is a continuing decision, not a signature stored from four years ago. Participants should know facts that could reasonably affect whether they stay. Here the safe fact is that an independent board reviewed the trial and recommended continuing. Releasing the unblinded interim numbers would let treatment knowledge spread back through participants and sites. Telling somebody their own arm would do the same. Saying nothing also fails the obligation, because a

meaningful review happened. The update should preserve both informed participation and the comparison still under way.

TAKEAWAY 6W

Consent is maintained, not collected once.

15.4 Why the board looks and the team does not

SAFETY MONITORING · MONITORING BOARD ROOM · A ROOM · CALLBACK · CHOICE

SCENE 34W

Vogt is in the building for an hour. Petrescu's closed report is on the table, face down. The medical director is copied on everything the trial team writes and on nothing the board writes.

GUIDE 54W

Four options, and somebody has to see the arms or a harmful treatment continues unnoticed. Ask of each whether it names a route from knowledge into the trial's own conduct. The board does not recruit, treat or score. The team does all three. Confidentiality cannot close those routes, and being advisory does not either.

ASK 14W

Why is the monitoring board shown the arms when the trial team is not?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

the trial team recruits, treats and measures while the trial runs

VERDICT 75W

Somebody must see unblinded data or a harmful treatment could continue unnoticed. The safeguard is separation of roles. The monitoring board does not recruit participants, deliver treatment or score outcomes. What it learns cannot change those processes directly. The trial team does all three. If that team knows which arm is winning, ordinary judgement can shift enrolment or follow-up. It can also shift how an ambiguous score is written. Confidentiality alone cannot remove those pathways.

TAKEAWAY 16W

The people who could act on an early result are the people kept from seeing it.

END OF THE DAY 17W

The last decision in a trial is what to say, and the numbers do not settle it.

THE CLOSING CARD

The board recommended continuing to the planned 380 events, with the investigator-rated symptom score dropped from the primary reporting and the four liver cases followed under a formal protocol. CLARION-3 reported eleven months later on 383 adjudicated events: an absolute reduction of 3.6 percentage points, about twenty-eight people treated for one event avoided, and an interval that excludes no benefit but is wider than anybody wanted.

What it cost: the symptom score, which was the outcome participants cared most about and the one the trial could no longer defend; a year in which half of those recruited went on receiving the arm that lost; and a subgroup finding in the over-seventies that two other groups have since failed to reproduce. What is unfinished: nobody knows what the drug does past two years, the liver signal is four cases and still four cases, and the next trial of this drug will have to be blinded some other way, because this one never really was.

Card text only — no options, boards or answers. 13,709 words of card prose across 61 stops. Generated from `themes/the_trial` by `tools/make-cardbook.mjs`.